INCH-POUND
{PD_NUMBER}
CAGE 98752
{PD_DATE}

PURCHASE DESCRIPTION

BEAR WATER SYSTEM FREEZE PROTECTION

1. SCOPE

1.1 Scope. This Purchase Description (PD) describes six modular and scalable Freeze Protection Modules (FPM) for the Basic Expeditionary Airfield Resources (BEAR) water system components.

2. APPLICABLE DOCUMENTS

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-C-490H	Chemical Conversion Coatings and Pretreatments for Metallic Substrates (Base for Organic Coatings)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5541F NOT 1	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-A-8625F(2) NOT 1	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-DTL-53022F(1)	Primer, Epoxy Coating, Corrosion Inhibiting Lead and Chromate Free
MIL-DTL-53030D(1) NOT 1	Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free

Comments, suggestions, or questions on this document should be addressed to: AFLCMC/ROZEC, 236 Milledgeville Street, Suite F30, Robins AFB GA 31098-1813. **Delete the following sentence if this is to be a PD:** Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC {FSC}

DISTRIBUTION STATEMENT A. Approved for public release.

MIL-DTL-81706B(1) NOT 1	Chemical Conversion Materials for Coating Aluminum and Aluminum Alloys
MIL-PRF-85285F	Coating: Polyurethane, Aircraft and Support Equipment

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130N(1) NOT 1	Identification Marking of U.S. Military Property
MIL-STD-461G	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-810H(1)	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-882E NOT 1	System Safety
MIL-STD-889D	Galvanic Compatibility of Electrically Conductive Materials
MIL-STD-1472H	Human Engineering

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

TECHNICAL ORDERS

40W4-21-11	Basic Expeditionary Airfield Resources (BEAR) Water System (DISTRO C)
40W4-20-1	Reverse Osmosis Water Purification Unit (ROWPU) (DISTRO C)
35E7-3-4-1	Heater, 130K Multi-Fueled, Portable, Duct Type (DISTRO A)
35E4-164-1	Hygiene System (DISTRO B)
50D1-4-1	Self – Help Laundry (DISTRO C)
35E4-248-1	BEAR Portable Electric Kitchen, Type 1 (DISTRO C)
AF TTP 3-32.34V1	Civil Engineer Bare Base Development
AF TTP 3-32.34V4	Contingency Water System
AF TTP 3-32.34V5	Contingency Electrical Power Production and Distribution Systems
40H3-7-1	Heater, Water 400,000 BTU (WH-400) PN 111244 (DISTRO A)
35E7-4-27-1	Heater, Water, Liquid Fuel M-80, NSN: 4520-01-162-0385 (DISTRO A)

CODE OF FEDERAL REGULATIONS (CFR)

40CFR82	Protection of Stratospheric Ozone

(Copies of this document are available online at http://www.ecfr.gov/.)

2.3 Non-Government publications. The following documents of the exact revision listed below form a part of this document to the extent specified herein.

International Organization for Standardization (ISO)

668	Containers, Series 1 Freight, Classification, Dimensions, and Ratings
1161	Containers, Series 1 Freight, Corner Fittings, Specifications
6346	Freight Containers, Coding, Identification and Marking
10012	Measurement Management Systems - Requirements for Measurement Processes and Measuring Equipment-First Edition.
1496-1	Containers, Series 1 Freight, Specification and Testing, Part 1: General Cargo Containers for General Purposes

(Applications for copies should be addressed to ISO, Case postale 56, 1211 Geneva 20, Switzerland, Fax +41 22 734 10 79.)

AEROSPACE INDUSTRIES ASSOCIATION (AIA) / NATIONAL AEROSPACE STANDARD (NAS)

NASM24667	Screw, Cap, Socket Head, Flat Countersunk, 82°, Alloy Steel, UNC-3A

(Application for copies should be addressed to Aerospace Industries of Association of America, Inc., 1000 Wilson Blvd., Arlington VA 22209, http://www.aia-aerospace.org/.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME) INTERNATIONAL

B18.3-2012	Socket Cap, Shoulder, Set Screws, and Hex Keys (Inch Series)

(Application for copies should be addressed to ASME International, Two Park Ave., New York NY 10016, http://www.asme.org/.)

ASTM INTERNATIONAL (ASTM)

F835-20	Standard Specification for Alloy Steel Socket Button and Flat Countersunk Head Cap Screws
F1136/F1136M-11	Standard Specification for Zinc/Aluminum Corrosion Protective Coatings for Fasteners

(Application for copies should be addressed to ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken PA 19428-2959, http://www.astm.org/.)

NACE INTERNATIONAL

SSPC-SP10/NACE No.2	Near-White Metal Blast Cleaning
SSPC-Vis 1	Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning

(Applications for copies should be addressed to NACE International, 15835 Park Ten Pl, Houston TX 77084, http://www.nace.org/.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	National Electrical Code, 2020

(Application for copies should be addressed to NFPA, 1 Batterymarch Park, Quincy MA 02169, http://www.nfpa.org/.)

SAE INTERNATIONAL

AMS-STD-595A	Colors Used in Government Procurement
J429-2014	Mechanical and Material Requirements for Externally Threaded Fasteners
J995-2017	Mechanical and Material Requirements for Steel Nuts

(Application for copies should be addressed to SAE International Customer Service, 400 Commonwealth Drive, Warrendale PA 15096, http://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Preproduction. When Specified (see 6.2), a sample kit or component shall be subjected to preproduction inspection in accordance with 4.2.

3.2 Description and general layout. The Basic Expeditionary Airfield Resources (BEAR) Water System is modular and scalable in 550 personnel support increments up to 3,300 personnel to meet a variety of user deployment needs. The BEAR Water System can draw water from a natural source, purify, store, and distribute the water while maintaining sufficient pressure, quantity, and quality for an entire base. Depending on the selected modular configuration, the water system can supply 30 gallons of water per person per day (gpppd) for a 1100-person camp and maintain enough storage capacity to sustain the camp for four days with no new water introduction. The BEAR Water System consists of a series of surface laid piping, pumping, fluid control, and water storage equipment that comprise distinct subsystems as shown in Figure 1.

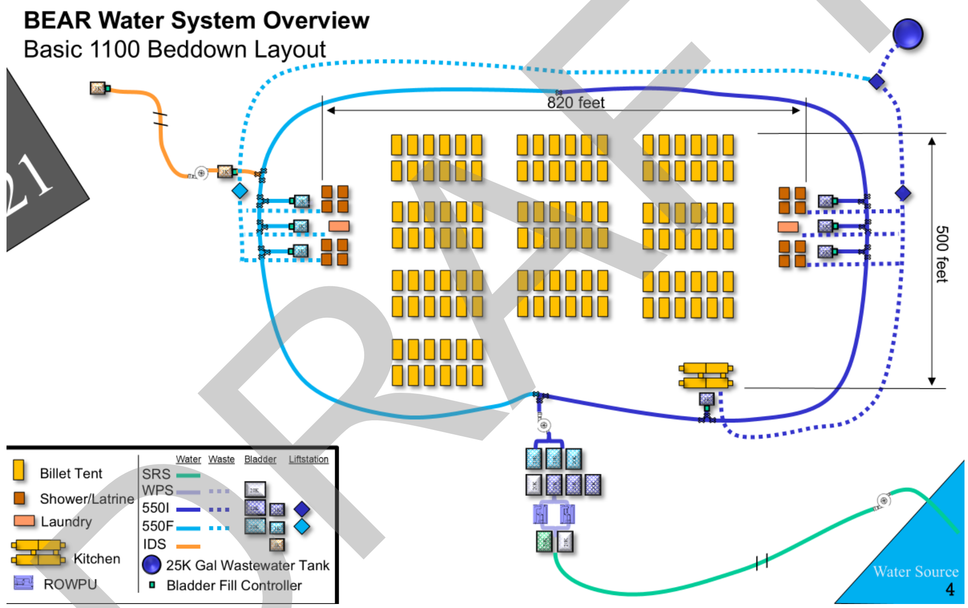


Figure 1: BEAR Water System Overview.

See Appendix I for BEAR Water System Major Component Details.

An overview of the following subsystems is provided in 3.2.1 through 3.2.5.

- Water Source Run
- Water Production
- Initial 550 Water Distribution (550I)
- Follow-On 550 Water Distribution (550F)
- Industrial Operations/Flightline Water Extension (IDS)

The following facility systems descriptions are provided in 3.2.6.

- Hygiene
- Kitchen
- Laundry

3.2.1 Water Source Run. The Source Run Subsystem (SRS) is a modular raw water distribution subsystem used to provide raw water from a source (pond, lake, stream, sea, or ocean) up to 6,300 feet to the water production subsystem. The SRS consists of a series of surface laid piping, plumbing, fluid control, and water storage equipment. These components include a 4 in. strainer and float assembly, 400 GPM diesel pump assembly, 20,000-gallon collapsible fabric tank, hoses, ball valves, and flaking boxes. Figure 2 provides an overview of the Water Source Run.

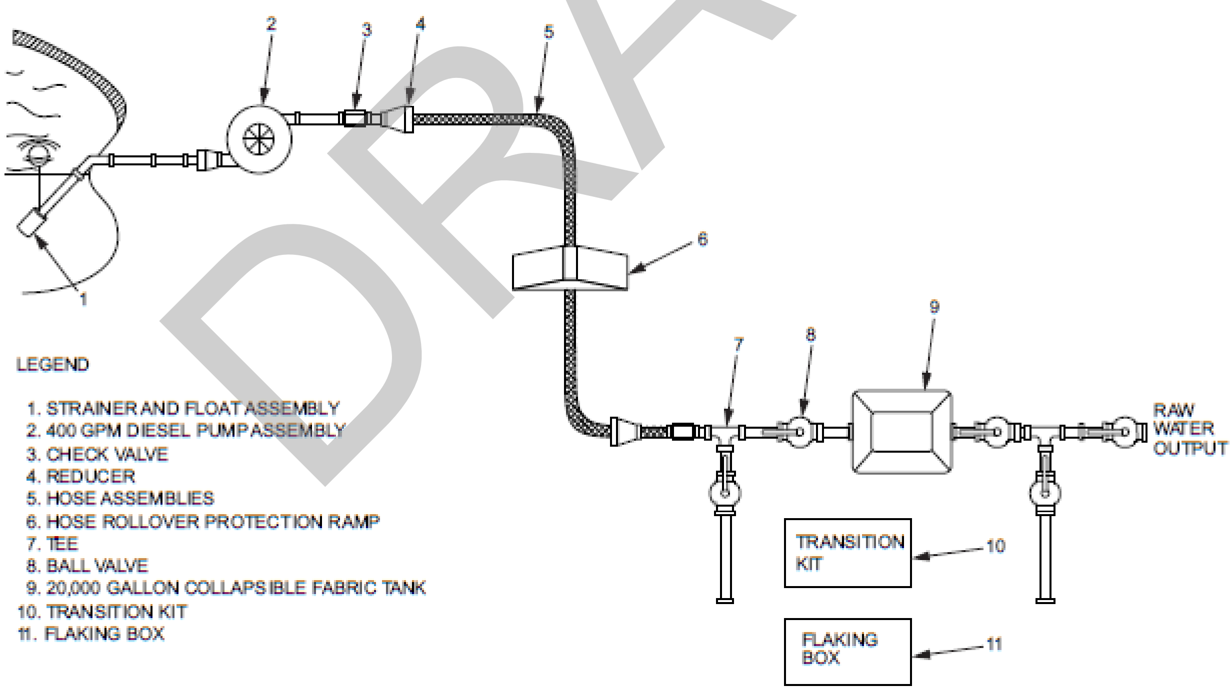


Figure 2: Water Source Run Subsystem.

3.2.2 Water Production. The Water Production Subsystem (WPS) generates potable water for distribution to user facilities within the BEAR water system. The WPS has the capability to receive from the SRS or draw directly from a source. Water purification is accomplished by two 1500 gallons per hour Reverse Osmosis Water Purification Units (ROWPUs). Raw water input is distributed through a 125 gallon per minute diesel pump for storage in a 20,000-gallon storage tank then distributed through a series of hoses and fittings to interface with the ROWPUs. The potable water generated by the ROWPUs is distributed through another series of surface laid hoses and fittings configured to interface with a 20,000-gallon potable water storage tank. Byproducts from the water purification process, brine water and wastewater, are distributed to a 20,000-gallon storage tank. The brine/wastewater storage tank is interfaced to a 35 gallon per minute electric pump for distribution to a waste area. The WPS consists of 2 each, 1500-gallon per hour ROWPUs and associated equipment. The equipment includes three 20,000-gallon bladders (Raw, waste, and potable water), 125 gallon per minute diesel pumps, electric pumps, various size hoses, valve assemblies, and associated fittings. Figure 3 provides an overview of the WPS.

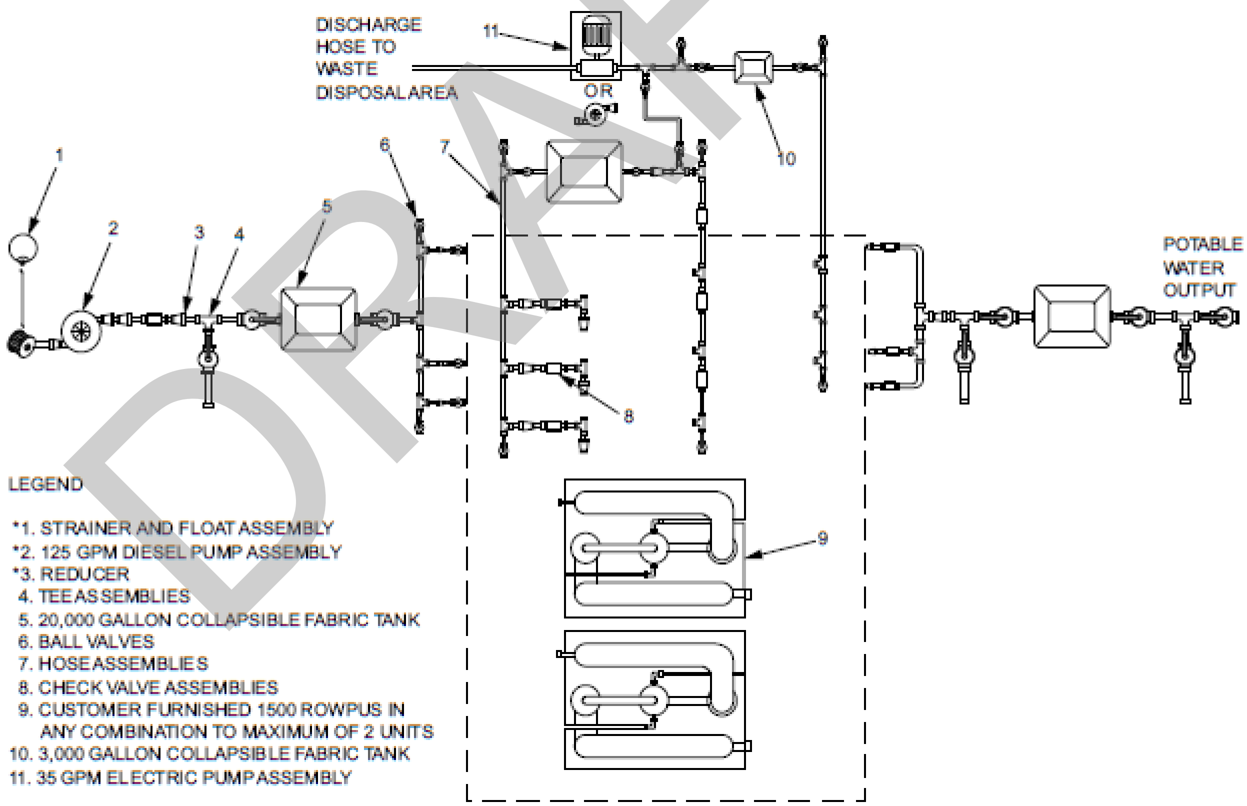


Figure 3: Water Production Subsystem.

3.2.3 Water Distribution Initial. The 550 Initial Subsystem (550I) is the primary potable water distribution system and contains the required surface laid components to distribute water on the Base to locations for laundry, hygiene, and kitchen to support a population of 550 personnel. The potable water input is distributed to three 20,000-gallon collapsible fabric tanks for storage, in a tank farm configuration. Water is distributed to a variable speed dual pumping station. The pumping station consists of two pumps configured in parallel. Output from the pumping station is applied to a distribution line that is looped and connected to form a pressurized feed line. The system can support up to a 6,000 feet distribution loop with a supply rating of 300 gallons per minute at 45 psi. User facilities are branched from the feed line. The 550I consists of adapters, 125 gallon per minute diesel pump, three 20,000-gallon collapsible fabric tanks, a dual pump station, four 3,000-gallon collapsible fabric tanks, four Bladder Water Level Controllers, two Sewage Ejector Systems, one Macerator Pump Lift Station, one Dual Pump Lift Station, one 25,000-gallon wastewater tank with mixing system, hoses and ball valves. Figure 4 provides an overview of the 550I subsystem.

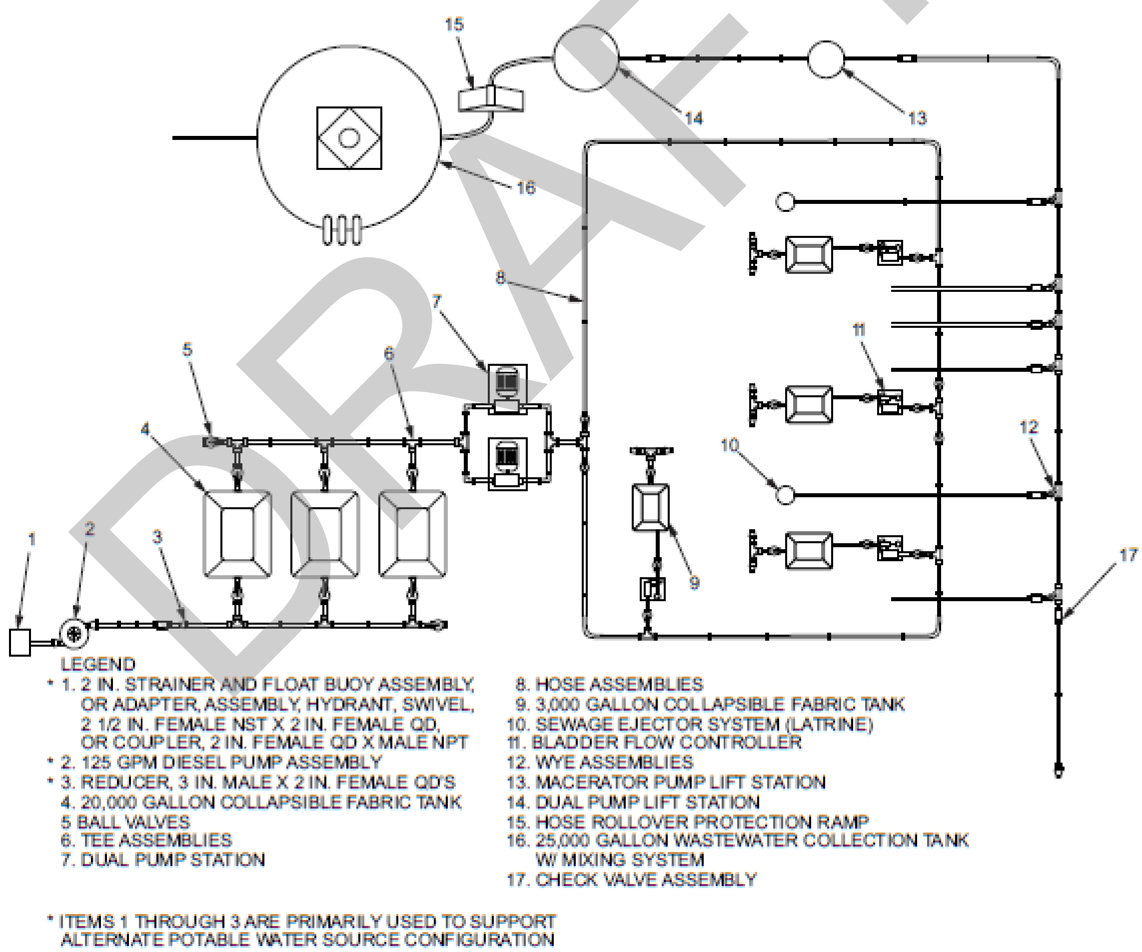


Figure 4: 550 Initial Subsystem.

3.2.4 Water Distribution Follow-On. The 550 Follow-On (550F) Subsystem provides for expansion and builds upon the 550I to support a population of 1100. The 550 Follow-On system is identical to the 550I, but does not provide a dual pump station, dual pump sewage lift station or 25,000-gallon wastewater collection tank. The 550F cannot be used independently. The 550F provides three 20,000-gallon collapsible fabric tanks, four 3,000-gallon collapsible fabric tanks, four Bladder Water Level Controllers, two Sewage Ejector Systems, one Sewage Ejector Lift Station, one Macerator Pump Station, hoses and ball valves. Figure 5 provides an overview of the 550F Subsystem.

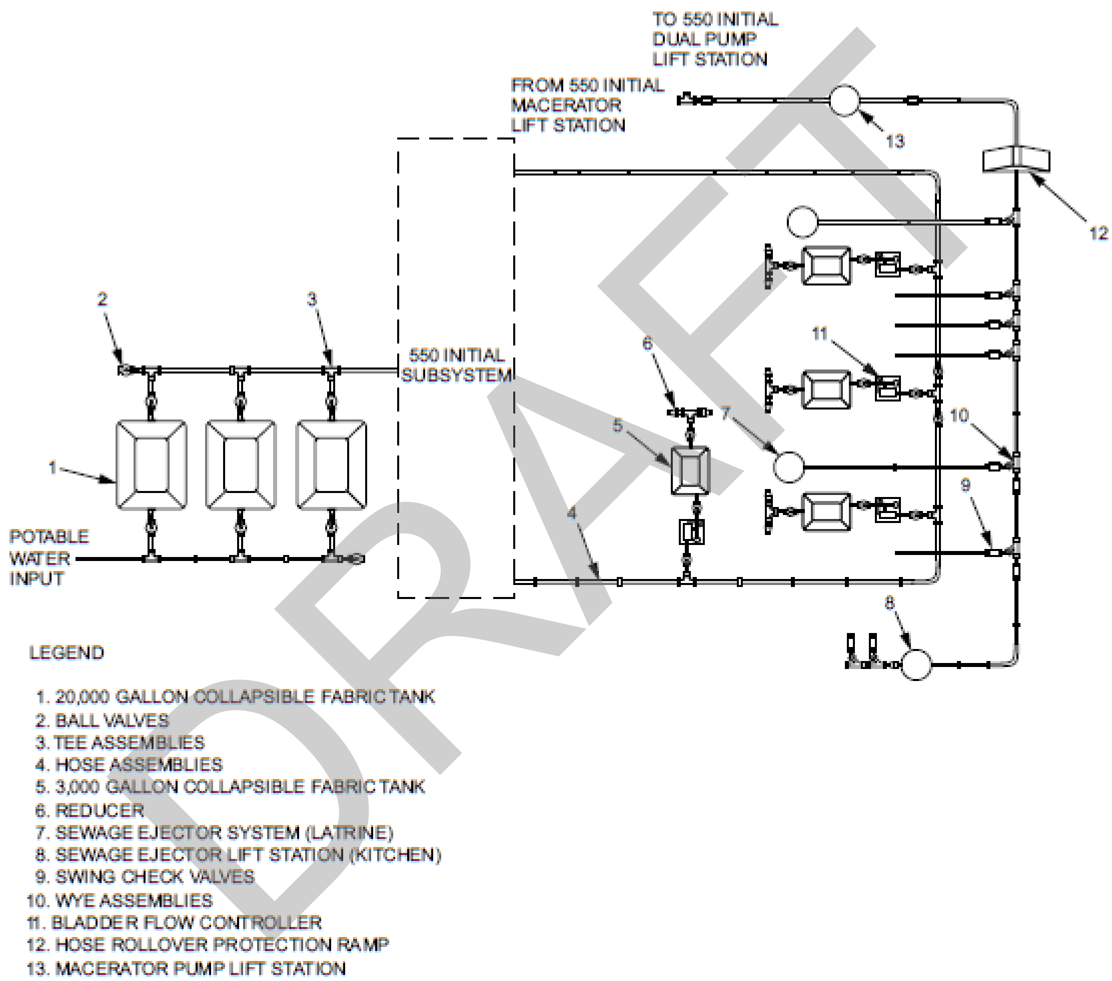
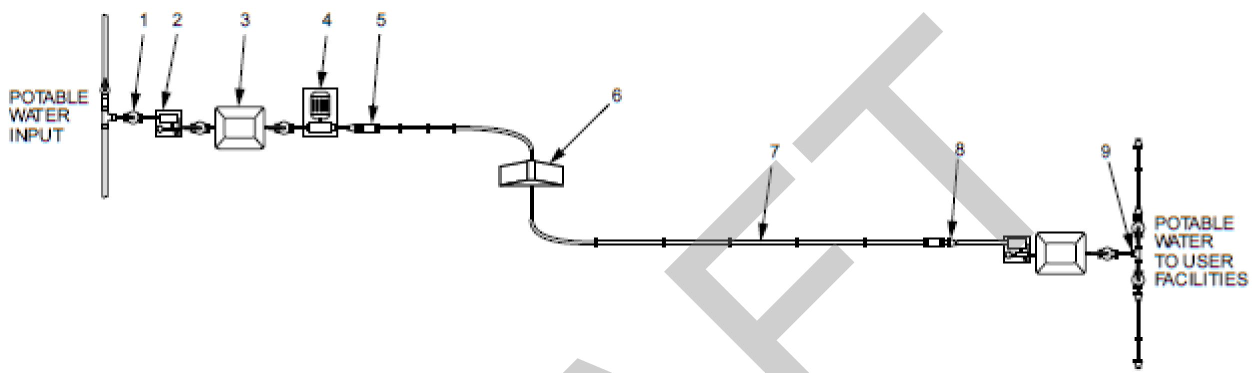


Figure 5: 550 Follow-On Subsystem.

3.2.5 Industrial Operations/Flightline Water Extension. The Industrial Operations Water Extension Subsystem provides surface laid potable water extension from the 550I, 550F, or Water Production subsystem by connection to any three-inch pressurized feed line. The system provides potable water to isolated user facilities up to 2,500 feet away. The Industrial Operations Water Extension Subsystem provides two bladder water level controllers, two 3,000-gallon collapsible fabric tanks, one 35 GPM electric pump, hoses, reducers, tee assemblies, ball valves, and check valves. Figure 6 provides an overview of the Flightline Water Extension.



LEGEND

1. BALL VALVES
2. BLADDER WATER LEVEL CONTROL
3. 3,000 GALLON COLLAPSIBLE FABRIC TANK
4. 35 GPM ELECTRIC PUMP ASSEMBLY
5. SWING CHECK VALVES
6. HOSE ROLLOVER PROTECTION RAMP
7. HOSE ASSEMBLIES
8. REDUCER
9. TEE ASSEMBLIES

Figure 6: Industrial Operations Water Extension Subsystem.

3.2.6 Facilities. The facilities systems include the expandable BICON Hygiene System, Kitchen and Laundry. The following systems are designed to interface with the BEAR water subsystems providing water and wastewater service.

3.2.6.1 Hygiene. The Hygiene System includes four expandable BICON shelters; consisting of two latrine BICON shelters and two shower BICON shelters. The Hygiene System utilizes various hoses, pipes, fittings, electrical pumps, and a AWH400 Fuel-burning water heater. Figure 7 provides an overview of the Hygiene system.

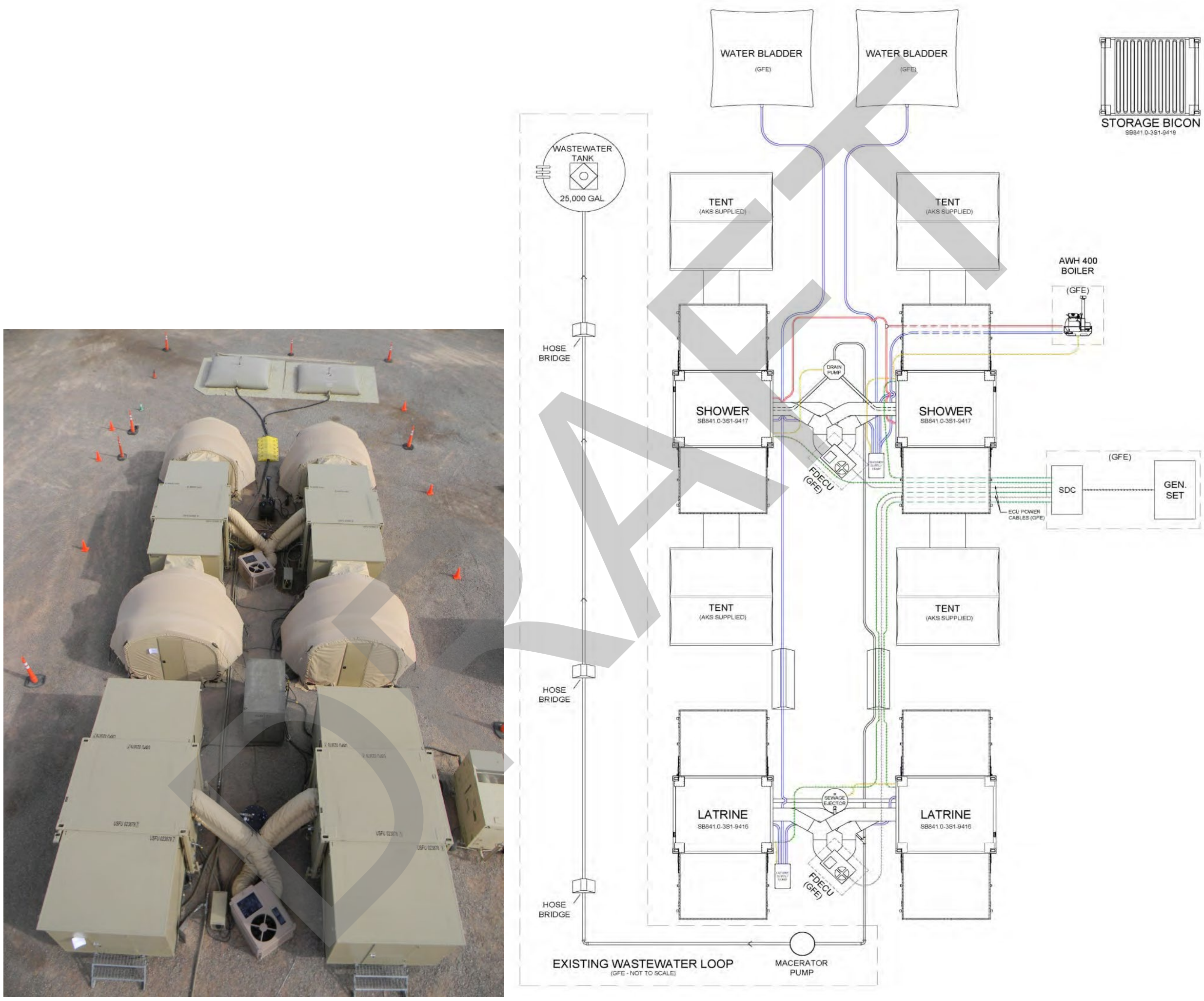
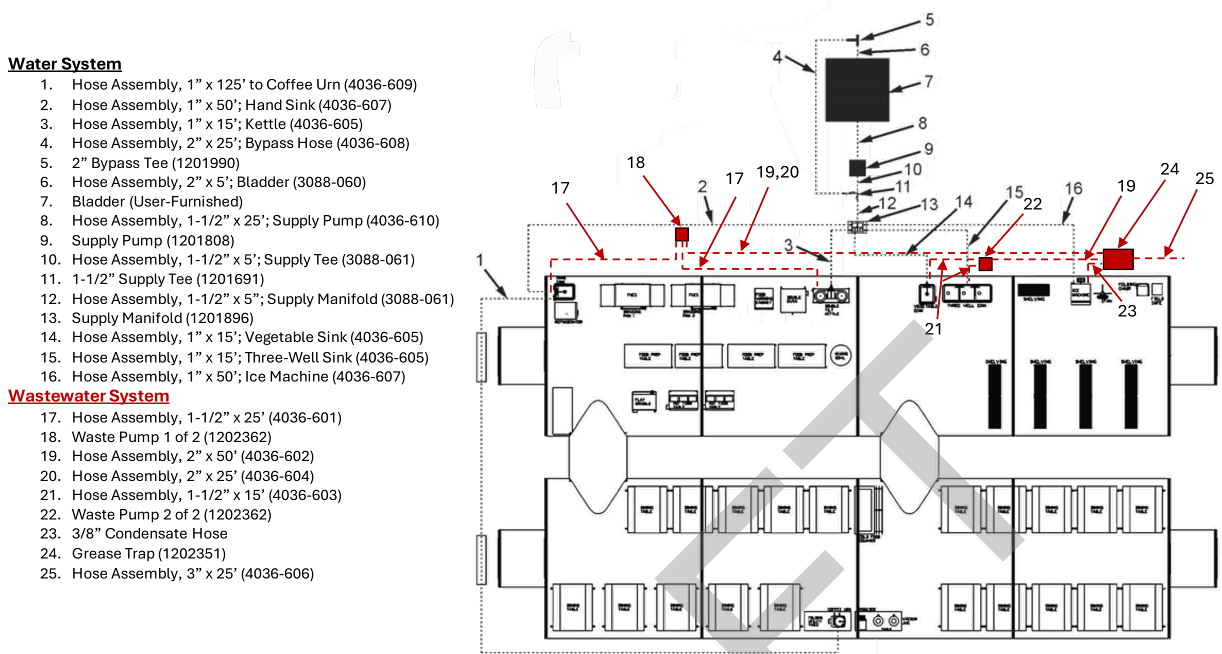


Figure 7: BICON Hygiene System.

3.2.6.2 Kitchen. The BEAR Type I Portable Electric Kitchen (PEK) provides mess services to



support up to 550 people. The PEK system includes complexed shelters, a complete meal preparation serving area, and a dining area. The PEK system utilizes various hoses, pipes, fittings, electrical pumps, a water bladder, and a grease trap for water distribution and waste collection. Figure 8 provides an overview of the PEK system.

Figure 8: Portable Electric Kitchen System.

3.2.6.3 Laundry. The Self-Help Laundry is an expeditionary self-service laundry mat housed within a single 32ft by 20ft shelter. The laundry includes five top loading washers, a water system with a 3,000-gallon water tank, water heater, supply pump, and drain pump.

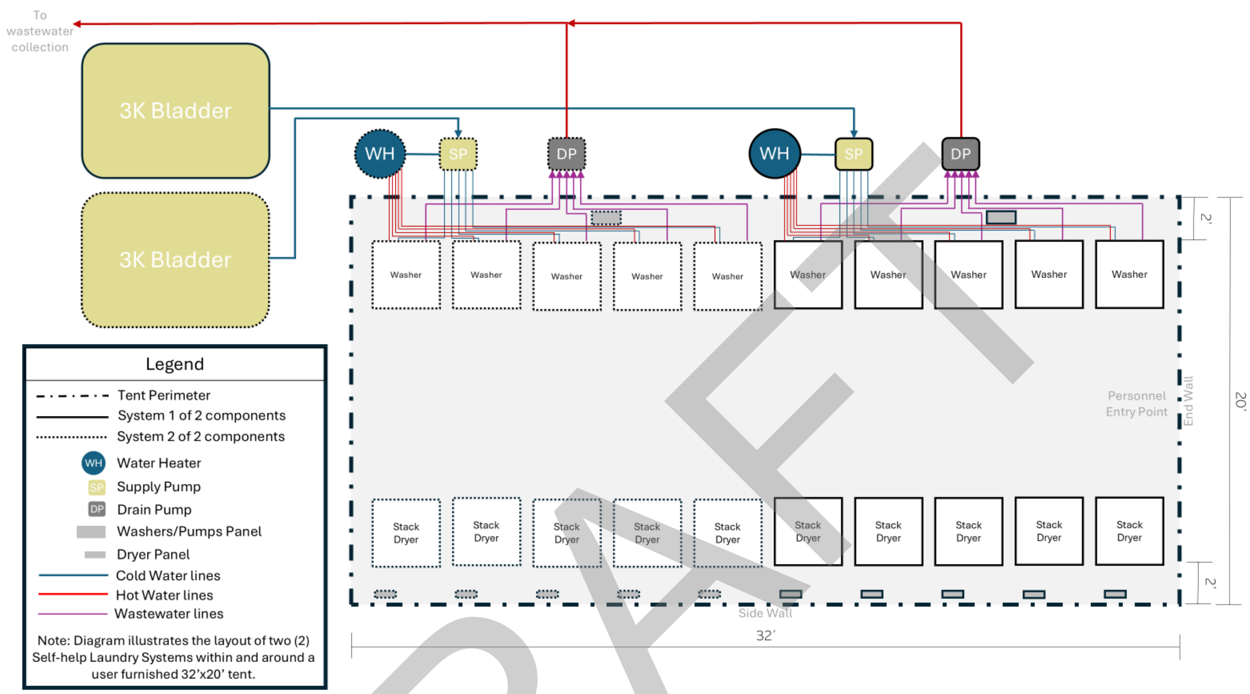


Figure 9: Self-Help Laundry.

3.3 BEAR Water Freeze Protection Overview. The Basic Expeditionary Airfield Resources (BEAR) Water Freeze Protection Modules (FPM) shall provide users with the equipment necessary to protect the BEAR water subsystem components from freezing. The BEAR FPM:

- Shall provide freeze protection of BEAR water system components in ambient temperature ranges from 32°F to -25°F (objective goal of -65°F).
- Shall provide the ability to automatically detect and provide visually indication of failed components or abnormal operational conditions, and may include interface for remote monitoring functionality.
- Shall have redundant capabilities and/or methods to overcome individual component failures.
- Shall provide the ability to return frozen system components to normal operating state.
- Shall provide the ability to penetrate ice layers for liquid water drawn from water source.

Any alternate technical approaches to requirements listed in this document shall be reviewed for approval by the procuring government activity.

The FPM shall be compatible with BEAR water distribution system without modification. Relocation of system components is acceptable; however, any additive component must interface with the current system. The FPM shall use items currently in the BEAR configuration to the maximum extent.

The FPM shall be modular and scalable in design to protect the water system pumping, piping, storage equipment; applicable fittings; and other system components. The modules shall be constructed of interchangeable pieces to maximize configuration flexibility. The modules shall be capable of indefinite scalability through the connection of additional modules. FPM shall not be functionally dependent upon any other FPM. The contractor shall maximize usage of commercial-off-the-shelf components and strive to minimize overall system weight without adversely affecting performance.

The FPM shall be designed and constructed to be transported, stored, assembled, disassembled, and operated at remote and austere locations in accordance with 3.7.

The FPMs shall provide modified layouts within Technical Manuals for the relocation of BEAR Laundry and Kitchen equipment. These layout modifications are detailed in section 3.

3.3.1 Module 1, Water Source Run Freeze Protection. FPM 1 shall provide protection through the identified component requirements in accordance with section 3.4. The module shall provide operators with the ability to penetrate ice layers for drafting water from a natural source and prevent inlet hose from freezing between operational cycles. All electrical components required to support freeze protection of the BEAR water system components shall be provided with the respective modules and support interfacing with the standard BEAR electrical distribution system as described in 3.4.2.

3.3.1.1 Freeze Protection for Water Source Run. FPM 1 shall provide heating and insulation for the 20k bladder to maintain water temperature above 50°F. The module shall provide insulated wrap with integrated heating to protect 1000 feet of 6-inch hose, 250 feet of 4-inch hose and associated valves and fittings. The module shall provide one 32ft. X 20ft. heated and insulated shelter to provide protection for the 400 GPM diesel pump, the shelter shall provide means to route motor exhaust out of shelter. Requirements for this module are described in Figure FPM1, and Table FPM1.

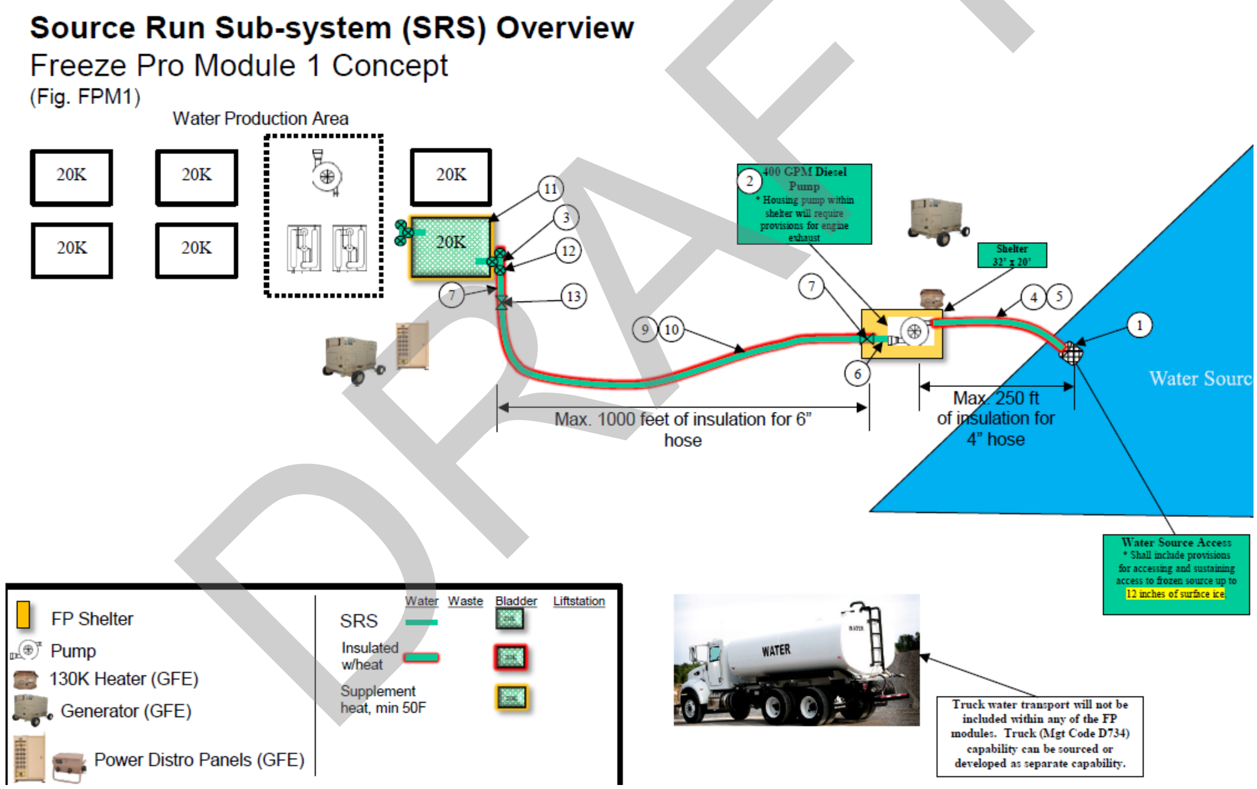


Figure FPM1: SRS Freeze Protection Module

Table FPM1: SRS Freeze Protection Module, Water System Components

Freeze Protection Module 1 (Subsystem(s): SRS)

Freeze Protection Requirements

FP Module	Subsystem Name	Water Sys Type	Item Category	Hose Dia.	NOMENCLATURE	Figure & Index/Sheet No.	Suggested FP Type	Total FPM Qty	FPM Length (ft)
FPM 1	**Source Run**	*Water System*	**EQUIP, WATER**		**STEEL SUCTION STRAINER, 4" MQD, w/Bouy Float**	FPM1-1	Special	1	0
					PUMP ASSY, DIESEL, 400 GPM, 4" FQD x MQD, GREEN STRIPE	FPM1-2	Shelter	1	0
			FITTING		**TEE, 4" FQD X MQD X MQD, GREEN STRIPE**	FPM1-3	Insulate	3	0
			HOSE	4"	**HOSE, DISCHARGE, 4" X 25', GREEN STRIPE**	FPM1-4	Insulate	2	50
					HOSE, DISCHARGE, 4" X 50', GREEN STRIPE	FPM1-5	Insulate	2	100
					HOSE, SUCTION, 4" X 15', GREEN STRIPE	FPM1-6	Insulate	4	60
					HOSE, SUCTION, 4" X 25', GREEN STRIPE	FPM1-7	Insulate	1	25
					HOSE, SUCTION, 4" X 50', GREEN STRIPE	FPM1-8	Insulate	1	50
				6"	**HOSE, DISCHARGE, 6" X 50', GREEN STRIPE**	FPM1-9	Insulate	2	100
					HOSE, DISCHARGE, 6" X 150', GREEN STRIPE	FPM1-10	Insulate	6	900
			TANK		**BLADDER, 20K GAL, GREEN STRIPE**	FPM1-11	Insulate	1	0
			VALVE		**VALVE, BALL, 4", GREEN STRIPE**	FPM1-12	Insulate	6	0
					VALVE, CHECK, 4", GREEN STRIPE	FPM1-13	Insulate	2	0
					Water System Total			31	1285
FPM 1 Total								**31**	**1285**

3.3.2 Module 2, Water Production Freeze Protection. FPM 2 shall provide protection through the identified component requirements in accordance with section 3.4. The module shall provide operators with the ability to penetrate ice layers for drafting water from a natural source and prevent inlet hose from freezing between operational cycles. All electrical components required to support freeze protection of the BEAR water system components shall be provided with the respective modules and support interfacing with the standard BEAR electrical distribution system as described in 3.4.2.

3.3.2.1 Freeze Protection for Water Production. FPM 2 shall provide one heated and insulated shelter with minimum dimensions of 50 ft. X 32ft. to provide protection for the ROWPUs, 3K water bladder, and electric pumps. FPM 2 shall provide dedicated freeze protection for the two diesel driven water pumps. Insulated hose wrap with integrated heating shall be provided for all exposed or unsheltered hoses and fittings. Requirements for this module are described in Figure FPM2, and Table FPM2.

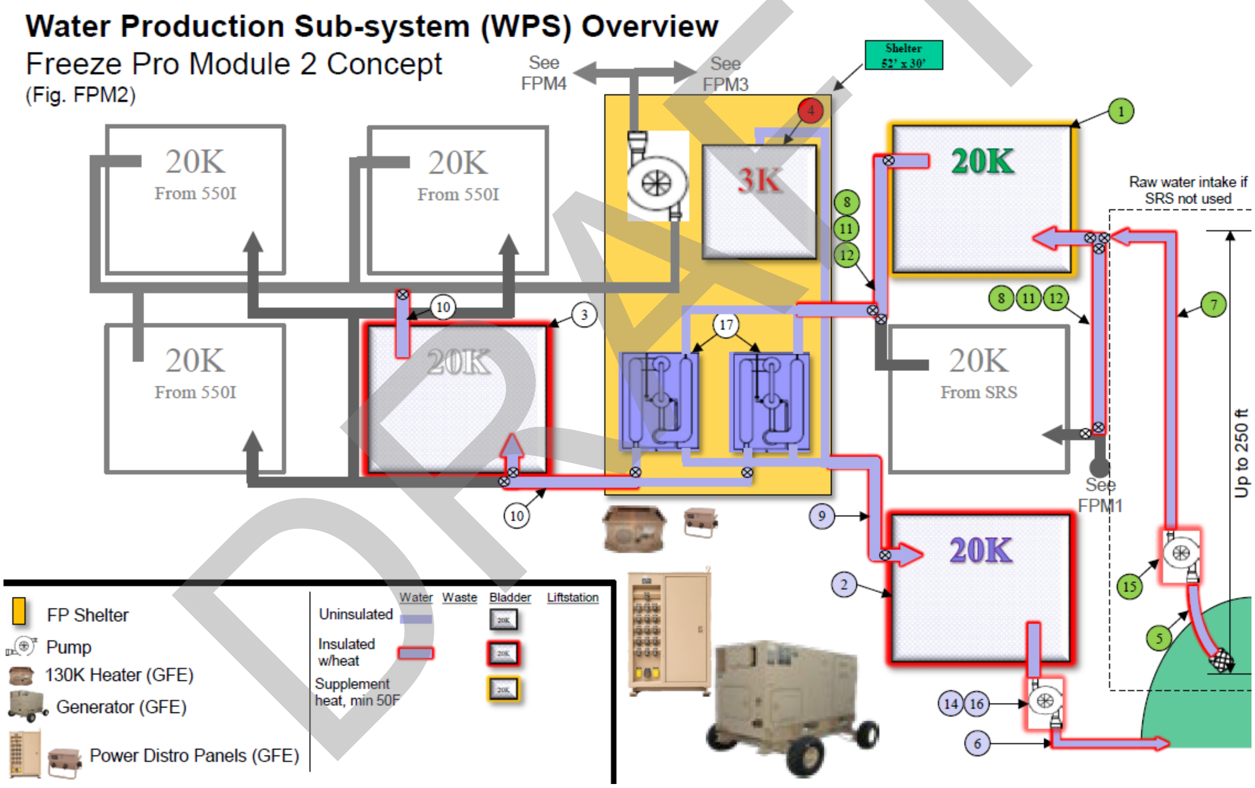


Figure FPM2: WPS Freeze Protection Module.

Table FPM2: WPS Freeze Protection Module, Water System Components

Freeze Protection Module 2 (Subsystem: WPS)

Freeze Protection Requirements

FP Module	Subsystem Name	Water Sys Type	Item Type	Hose Dia.	NOMENCLATURE	Figure & Index/Sheet No.	Suggested FP Type	Total FPM Qty	FPM Length (ft)
FPM 2	Water Production	*Water System*	BLADDER		BLADDER, 20K GAL, GREEN STRIPE	FPM2-1	Insulate	1	0
					BLADDER, 20K GAL, PURPLE STRIPE	FPM2-2	Insulate	1	0
					BLADDER, 20K GAL, WHITE STRIPE	FPM2-3	Insulate	1	0
					BLADDER, 3K GAL, RED STRIPE	FPM2-4	Insulate	1	0
			HOSE	2"	HOSE, SUCTION, 2" X 25', GREEN STRIPE	FPM2-5	Insulate	2	50
				3"	HOSE, DISCHARGE, 3" X 100', PURPLE STRIPE	FPM2-6	Insulate	3	300
					HOSE, DISCHARGE, 3" X 100', GREEN STRIPE	FPM2-7	Insulate	3	300
				4"	HOSE, SUCTION, 4" X 15' GREEN STRIPE	FPM2-8	Insulate	8	120
					HOSE, SUCTION, 4" X 15' PURPLE STRIPE	FPM2-9	Insulate	5	75
					HOSE, SUCTION, 4" X 15' WHITE STRIPE	FPM2-10	Insulate	8	120
					HOSE, SUCTION, 4" X 25', GREEN STRIPE	FPM2-11	Insulate	1	25
					HOSE, SUCTION, 4" X 50', GREEN STRIPE	FPM2-12	Insulate	2	100
					HOSE, SUCTION, 4" X 25', PURPLE STRIPE	FPM2-13	Insulate	4	100
			PUMP		PUMP ASSY, ELECTRIC, 2" FQD X MQD, PURPLE STRIPE	FPM2-14	Insulate	1	0
					PUMP ASSY, DIESEL, 125 GPM, 2" FQD x MQD, GREEN STRIPE	FPM2-15	Insulate	1	0
					PUMP ASSY, DIESEL, 125 GPM, 2" FQD x MQD, PURPLE STRIPE	FPM2-16	Insulate	1	0
			ROWPU		1500 ROWPU	FPM2-17	Shelter	2	0
			TEE		TEE, 4" MQD X MQD X FQD, GREEN STRIPE		Insulate	2	0
					TEE, 4" FQD X MQD X MQD, GREEN STRIPE		Insulate	7	0
					TEE, 4" FQD X MQD X FQD, PURPLE STRIPE		Insulate	4	0
					TEE, 4" FQD X MQD X MQD, PURPLE STRIPE		Insulate	2	0
					TEE, 4" FQD X MQD X MQD, WHITE STRIPE		Insulate	3	0
			VALVE, BALL		VALVE, BALL, 4", PURPLE STRIPE		Insulate	5	0
					VALVE, BALL, 4", WHITE STRIPE		Insulate	7	0
					VALVE, BALL, 4", GREEN STRIPE		Insulate	8	0
					Water System Total			*83*	*1190*

3.3.3 Module 3, Water Distribution Initial Freeze Protection. FPM 3 for 550I with integration of Hygiene, Kitchen, and Laundry systems shall provide freeze protection through the identified component requirements in accordance with section 3.4. All electrical components required to support freeze protection of the BEAR water system components shall be provided with the respective modules and support interfacing with the standard BEAR electrical distribution system as described in 3.4.2.

3.3.3.1 Freeze Protection for 550I. FPM 3 shall provide freeze protection for the potable water distribution portion of the 550I that includes 3 each - 20K pillow type water bladders with accessory hoses and fittings, 1,000 ft. of 3-inch water hoses, 400 ft. of 4-inch hose, 400 ft. of 2-inch hose, a dual pump station, and a 125GPM diesel pump. FPM 3 shall include a 32ft. X 20ft. heated and insulated shelter to house 3 each – 3K pillow type water bladders with bladder fill controller assemblies that provide water supply to the Hygiene, and Laundry systems.

FPM 3 shall provide freeze protection for the wastewater collection portion of the 550I that includes 1 each - 25K waste tank system, 1,000 ft. of 3 to 4-inch wastewater hose, 400 ft. of 1 ½ to 2-inch hose, a macerator pump lift station, and a dual pump lift station.

Requirements for this module are described in Figures FPM3, sheets 1 - 7, and Tables FPM3-550I Water System, FPM3-550I Wastewater System, FPM3-PEK, FPM3-Laundry, FPM3-Hygiene.

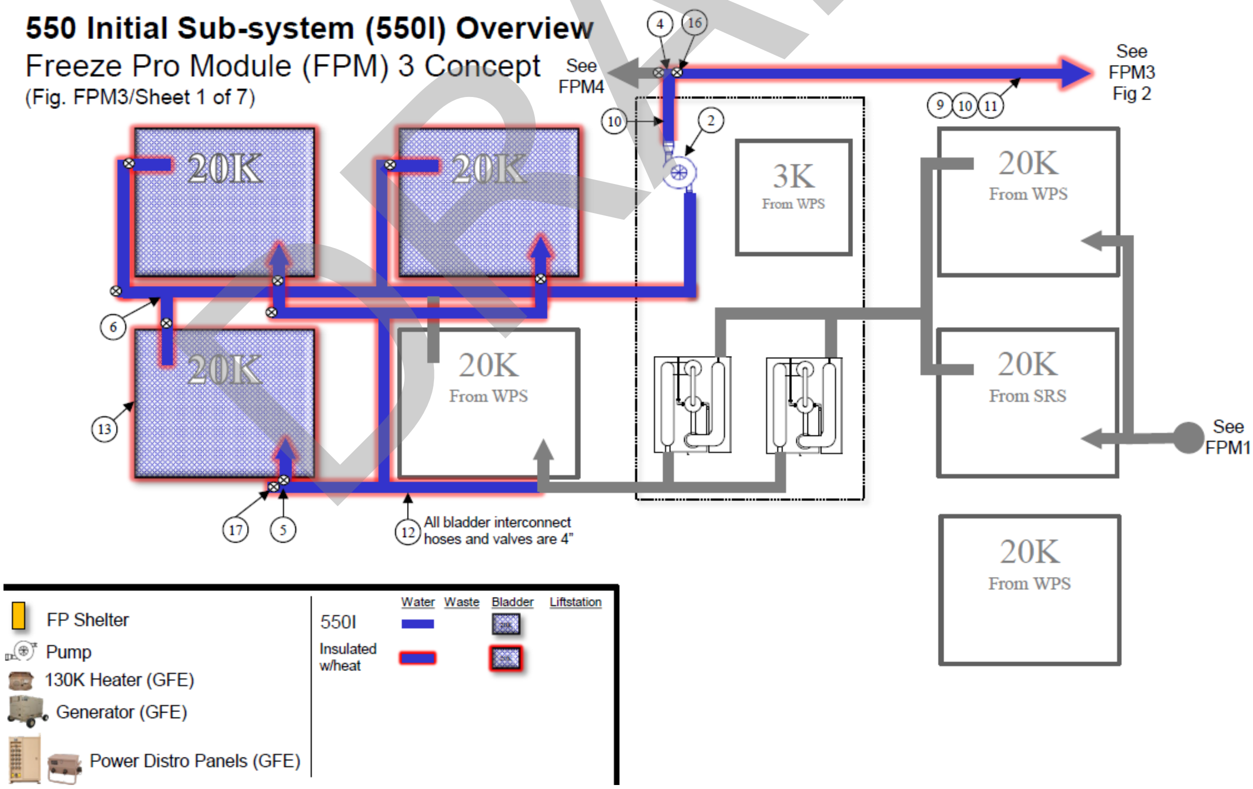


Figure FPM3, 1 of 7: 550I Freeze Protection.

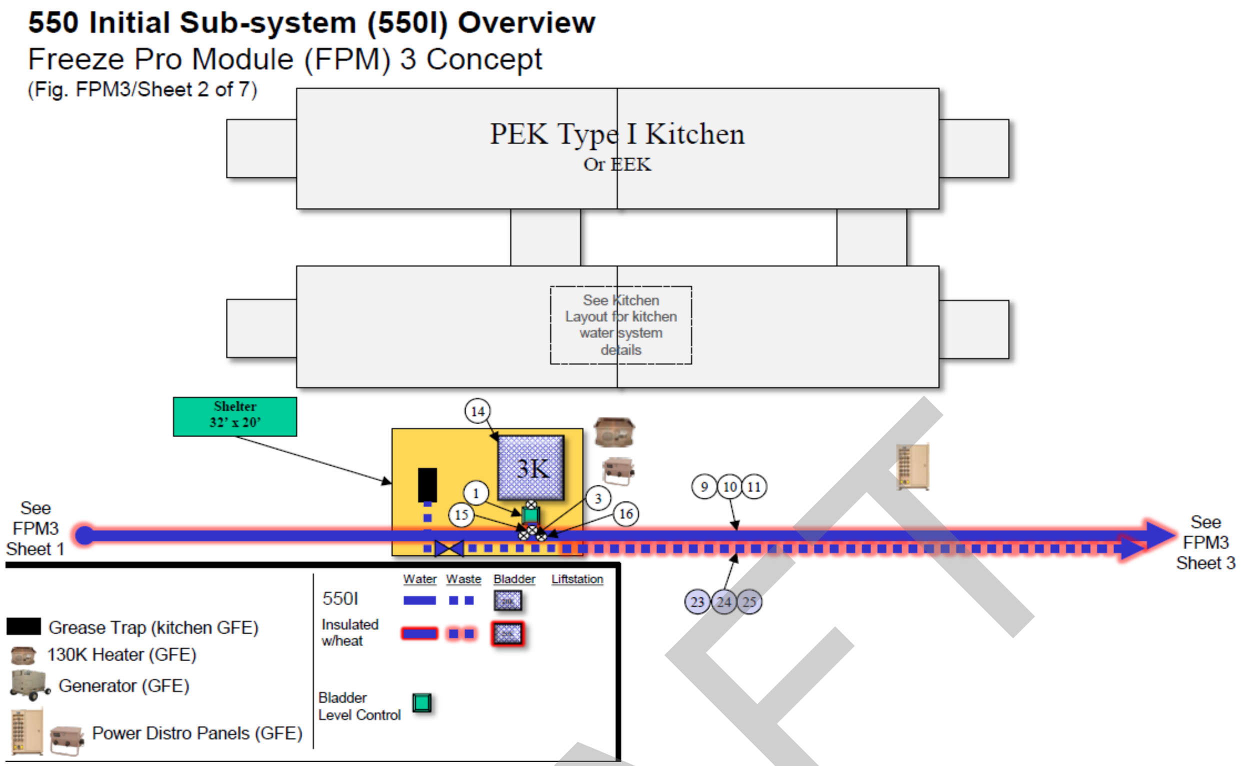


Figure FPM3, 2 of 7: 550I Freeze Protection Continued.

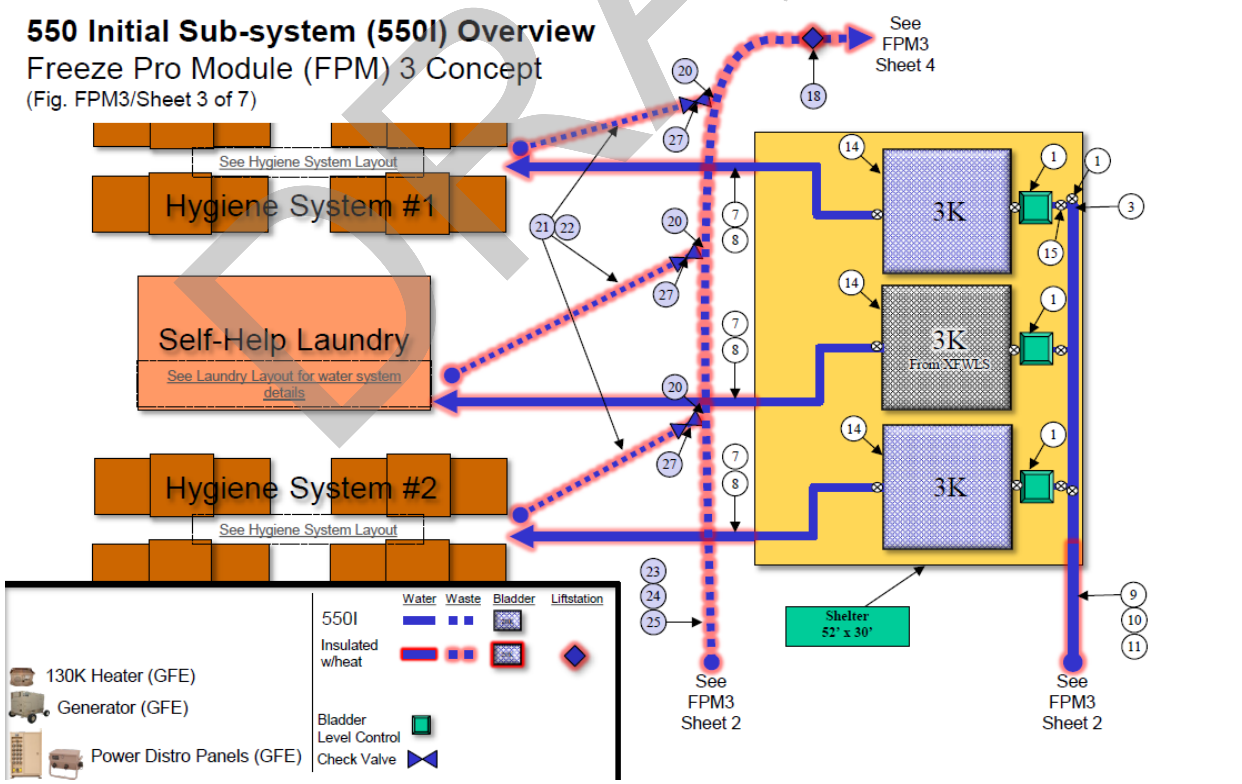


Figure FPM3, 3 of 7: 550I Freeze Protection Continued

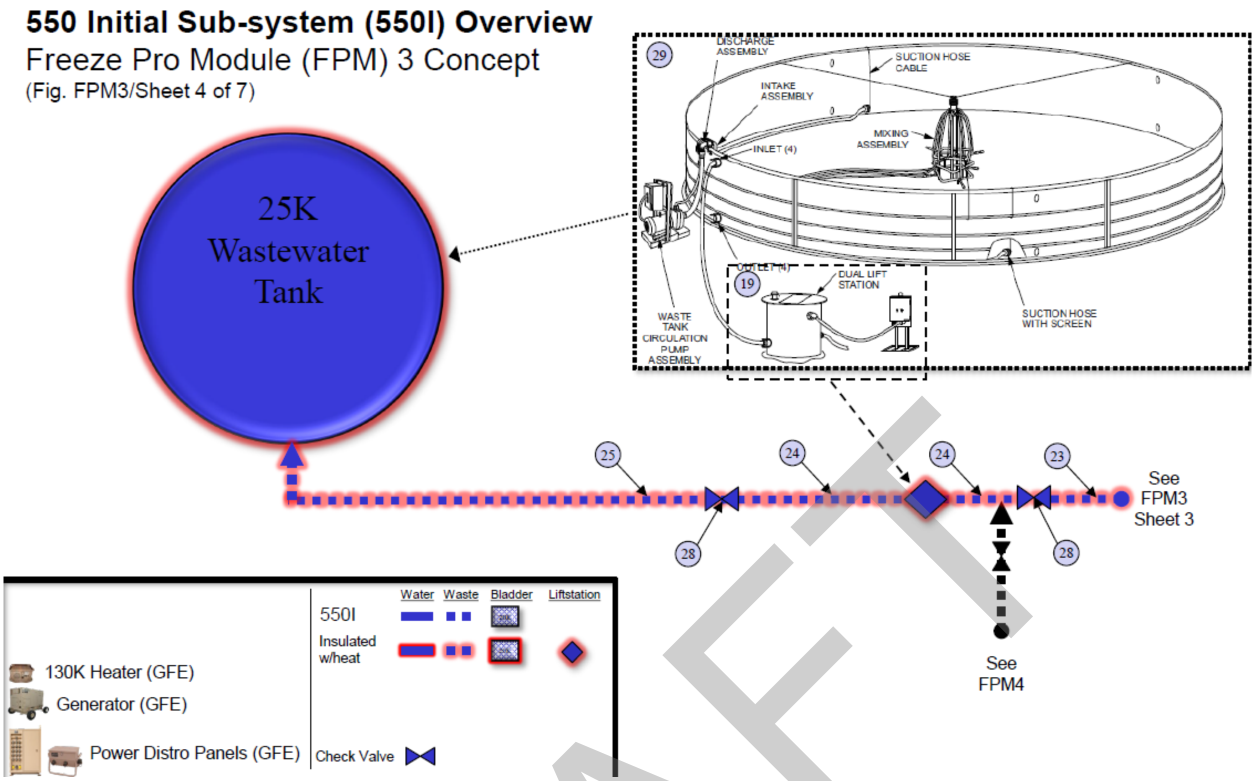


Figure FPM3, 4 of 7: 550I Freeze Protection Continued

Table FPM3: 550I Freeze Protection Module, Water System Components

Freeze Protection Module 3 (Subsystem: 550I)

Freeze Protection Requirements (Sheets 1-4)

FP Module	Subsystem Name	Water Sys Type	Item Category	Hose Dia.	NOMENCLATURE	Figure & Index/Sheet No.	Suggested FP Type	Total FPM Qty	*FPM Length (ft)
FPM 3	550I	*Water System*	EQUIP, WATER		CONTROL, BLADDER WATER LEVEL W/25' EX CORD	FPM3-1/2,3	Shelter	4	0
					DUAL PUMPING STATION	FPM3-2/1	Shelter	1	0
			FITTING		TEE, 3" FQD X 3" MQD X 2" MQD, WHITE STRIP	FPM3-3/2,3	Insulate	4	0
					TEE, 3" FQD X 3" MQD X 3" FQD, WHITE STRIPE	FPM3-4/1	Insulate	2	0
					TEE, 4" FQD X 4" MQD X 4" MQD, WHITE STRIPE	FPM3-5/1	Insulate	4	0
					TEE, 4" FQD X 4" MQD X 4" FQD, WHITE STRIPE	FPM3-6/1	Insulate	4	0
			HOSE	2"	HOSE, SUCTION, 2" X 25', WHITE STRIPE	FPM3-7/3	Insulate	6	150
					HOSE, SUCTION, 2" X 50', WHITE STRIPE	FPM3-8/3	Insulate	1	50
				3"	HOSE, DISCHARGE, 3" X 100', WHITE STRIPE	FPM3-9/1,2,3	Insulate	7	700
					HOSE, DISCHARGE, 3" X 25', WHITE STRIPE	FPM3-10/1,2,3	Insulate	4	100
					HOSE, DISCHARGE, 3" X 50', WHITE STRIPE	FPM3-11/1,2,3	Insulate	4	200
				4"	HOSE, SUCTION, 4" X 15', WHITE STRIPE	FPM3-12/1	Insulate	15	225
			TANK		BLADDER, 20K GAL, WHITE STRIPE	FPM3-13/1	Insulate	3	0
					BLADDER, 3K GAL, WHITE STRIPE	FPM3-14/2,3	Shelter	3	0
			VALVE		VALVE, BALL, 2", WHITE STRIPE	FPM3-15/2,3	Insulate	4	0
					VALVE, BALL, 3", WHITE STRIPE	FPM3-16/1,2,3	Insulate	4	0
					VALVE, BALL, 4", WHITE STRIPE	FPM3-17/1	Insulate	8	0
					Water System Total			*78*	*1425*

Table FPM3: 550I Freeze Protection Module, Wastewater System Components

Freeze Protection Module 3 (Subsystem: 550I)

Freeze Protection Requirements (Sheets 1-4)

FP Module	Subsystem Name	Water Sys Type	Item Category	Hose Dia.	NOMENCLATURE	Figure & Index/Sheet No.	Suggested FP Type	Total FPM Qty	FPM Length (ft)
FPM 3	550I	*Wastewater System*	EQUIP, WATER		MACERATOR PUMP SEWAGE LIFT STATION	FPM3-18/3	Insulate	1	0
					DUAL PUMP SEWAGE LIFT STATION	FPM3-19/4	Insulate	1	0
			FITTING		WYE, 3" FQD X 3" MQD X 2" FQD, PURPLE STRIPE	FPM3-20/3	Insulate	4	0
			HOSE	2"	HOSE, DISCHARGE, 2" X 25', PURPLE STRIPE	FPM3-21/3	Insulate	2	50
					HOSE, DISCHARGE, 2" X 50', PURPLE STRIPE	FPM3-22/3	Insulate	5	250
				3"	HOSE, DISCHARGE, 3" X 100', PURPLE STRIPE	FPM3-23/2,3,4	Insulate	5	500
					HOSE, DISCHARGE, 3" X 25', PURPLE STRIPE	FPM3-24/2,3,4	Insulate	5	125
					HOSE, DISCHARGE, 3" X 50', PURPLE STRIPE	FPM3-25/2,3,4	Insulate	4	200
			VALVE		VALVE, BALL, 4", PURPLE STRIPE	FPM26/3	Insulate	3	0
					VALVE, CHECK, 2", PURPLE STRIPE	FPM27/3	Insulate	4	0
					VALVE, CHECK, 3", PURPLE STRIPE	FPM28/3	Insulate	3	0
			25K WCT		25K GAL WASTE COLLECTION SYSTEM	FPM3-29/4			
				1-1/2"	HOSE, DISCHARGE, 1-1/2" X 15', PURPLE STRIPE (25K WCT)		Insulate	2	0
				2"	HOSE, DISCHARGE, 2" X 5', PURPLE STRIPE (25K WCT)		Insulate	1	0
				3"	HOSE, SUCTION, 3" X 30', PURPLE STRIPE (25K WCT)		Insulate	1	0
					HOSE, SUCTION, 3" X 5', PURPLE STRIPE (25K WCT)		Insulate	1	0
				4"	HOSE, DISCHARGE, 4" X 50', PURPLE STRIPE		Insulate	4	0
					25K WTC MIXING SYS INTAKE ASSY (J-TUBE) (25K WCT)		Insulate	1	0
					25K GALLON WASTE COLLECTION TANK (25K WCT)		Insulate	1	0
					25K WASTEWATER TANK MIXING SYS (25K WCT)		Insulate	1	0
					25K WASTE TANK AERATOR MIXING ASSEMBLY (25K WCT)		Insulate	1	0
					25K WASTE TANK MIXING PUMP ASSY (25K WCT)		Insulate	1	0
					25K WASTE TANK DISCHARGE ASSY (MANIFOLD) (25K WCT)		Insulate	1	0
					Wastewater System Total			*52*	*1125*

3.3.3.2 <u>Freeze Protection for Kitchen</u>. FPM 3 shall include a 32ft. X 20ft. heated and insulated shelter to house 1 each – 3K pillow type water bladder, electric water pump, and grease trap assembly for the Kitchen system.

All pumps, equipment and hoses will be relocated within the kitchen shelter as illustrated in figure FPM3 Sheet 5, except for exposed water supply lines between shelters. The exposed lines are 1 ½ - inch X 15ft. water hose and 2ea 2-inch X 15ft. wastewater discharge hose.

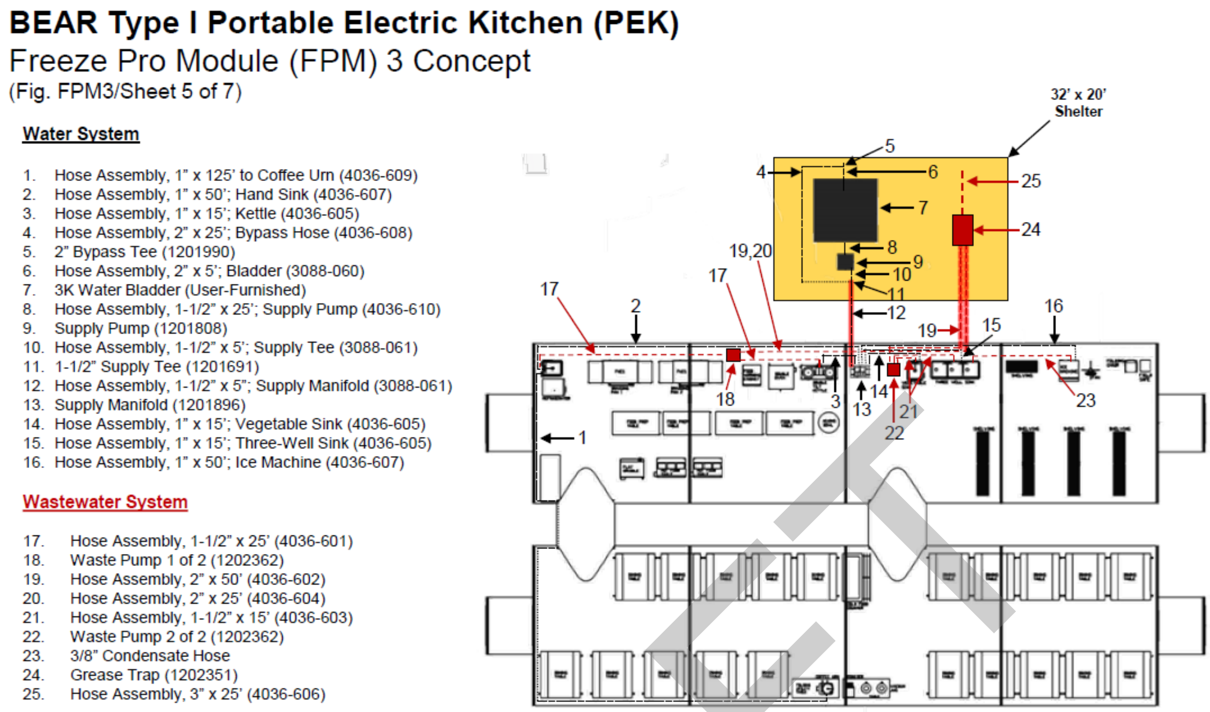


Figure FPM3, 5 of 7: PEK Freeze Protection, 550I Freeze Protection Continued.

Table FPM3-PEK: 550I Freeze Protection Module, PEK Components

Freeze Protection Module 3 (Subsystem: PEK)
Freeze Protection Requirements

FP Module	Subsystem Name	Water Sys Type	Item Category	Hose Dia.	NOMENCLATURE	Figure & Index/Sheet No.	Suggested FP Type	Total FPM Qty	*FPM Length (ft)
FPM 3	PEK	*Water System*	EQUIP, WATER		PUMP, SUPPLY	FPM3-9/5	Shelter	1	0
			HOSE	1-1/2"	HOSE, SUCTION, 1-1/2" X 25', SUPPLY	FPM3-12/5	Insulate	1	25
					Water System Total			2	25
		Wastewater System	EQUIP, WATER		GREASE TRAP	FPM3-24/5	Shelter	1	0
			HOSE	2"	HOSE, SUCTION, 2" X 50', WASTE	FPM3-19/5	Insulate	2	100
					Wastewater System Total			3	100

3.3.3.3 Freeze Protection for Laundry. The Laundry System Water Source, 3k bladder, is housed in the shelter identified in 3.3.3.1. All pumps, equipment and hoses will be relocated within the laundry shelter as illustrated in figure FPM3 Sheet 6, except for the exposed water supply lines between shelters and the wastewater discharge that extends beyond the laundry shelter. The FPM 3 shall provide freeze protection for 40ft. of 1 ½-inch water supply lines, 40ft. of 2-inch wastewater lines.

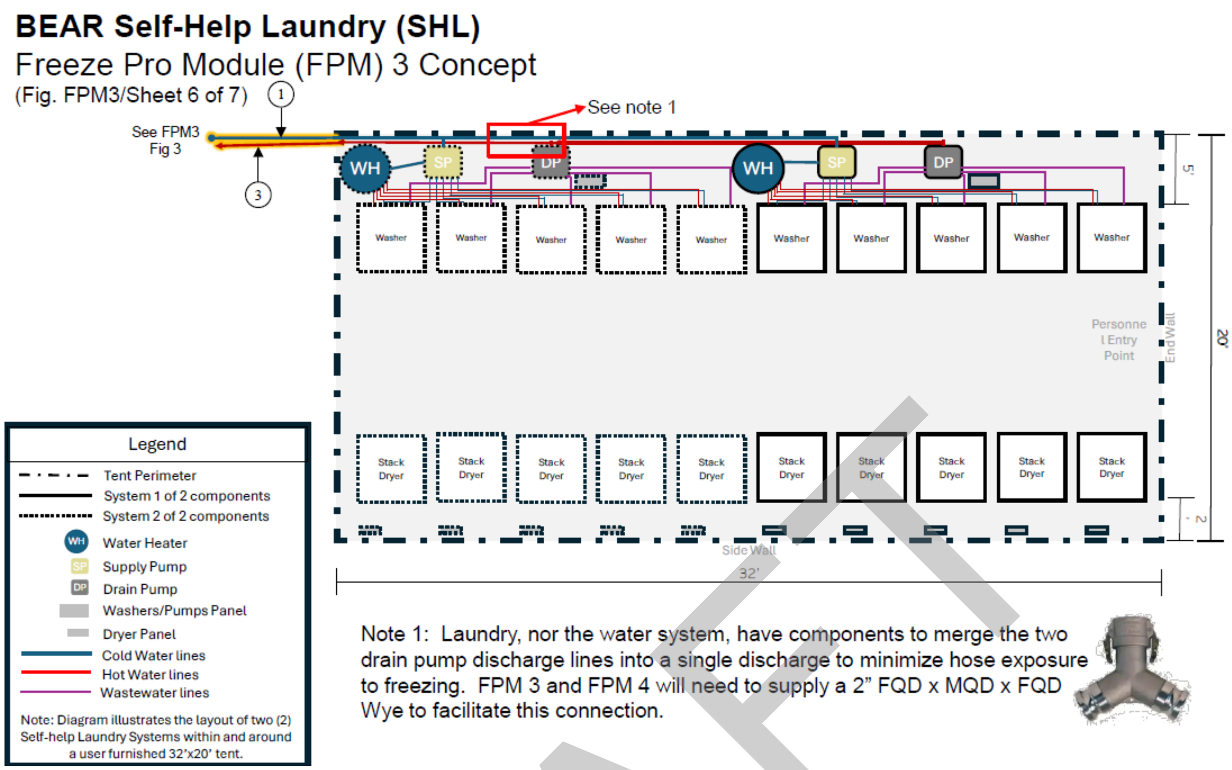


Figure FPM3, 6 of 7: Self-Help Laundry Freeze Protection, 550I Freeze Protection Continued.

Table FPM3-Laundry: 550I Freeze Protection Module, Self-Help Laundry Components

Freeze Protection Module 3 (Subsystem: Laundry)
Freeze Protection Requirements

FP Module	Subsystem Name	Water Sys Type	Item Category	Hose Dia.	NOMENCLATURE	Figure & Index/Sheet No.	Suggested FP Type	Total FPM Qty	*FPM Length (ft)
FPM 3	Self-Help Laundry	Water Sys	HOSE	1-1/2"	HOSE, SUCTION, 1-1/2" x 25'	FPM3-1/6	Insulate	2	50
			TANK		BLADDER, 3K GAL, WHITE STRIPE	FPM3-14/3	Shelter	1	0
						Water Sys Total		3	50
		Wastewater Sys	HOSE	2"	HOSE, DISCHARGE, 2" x 25'	FPM3-3/6	Insulate	2	50
						Wastewater Sys Total		2	50

3.3.3.4 Freeze Protection for Hygiene. The Hygiene Systems water source, 3K bladder, is housed in the shelter identified in 3.3.3.1. Requirements for this module are described in Figures FPM3 Sheet 7, and Table FPM3-Hygiene.

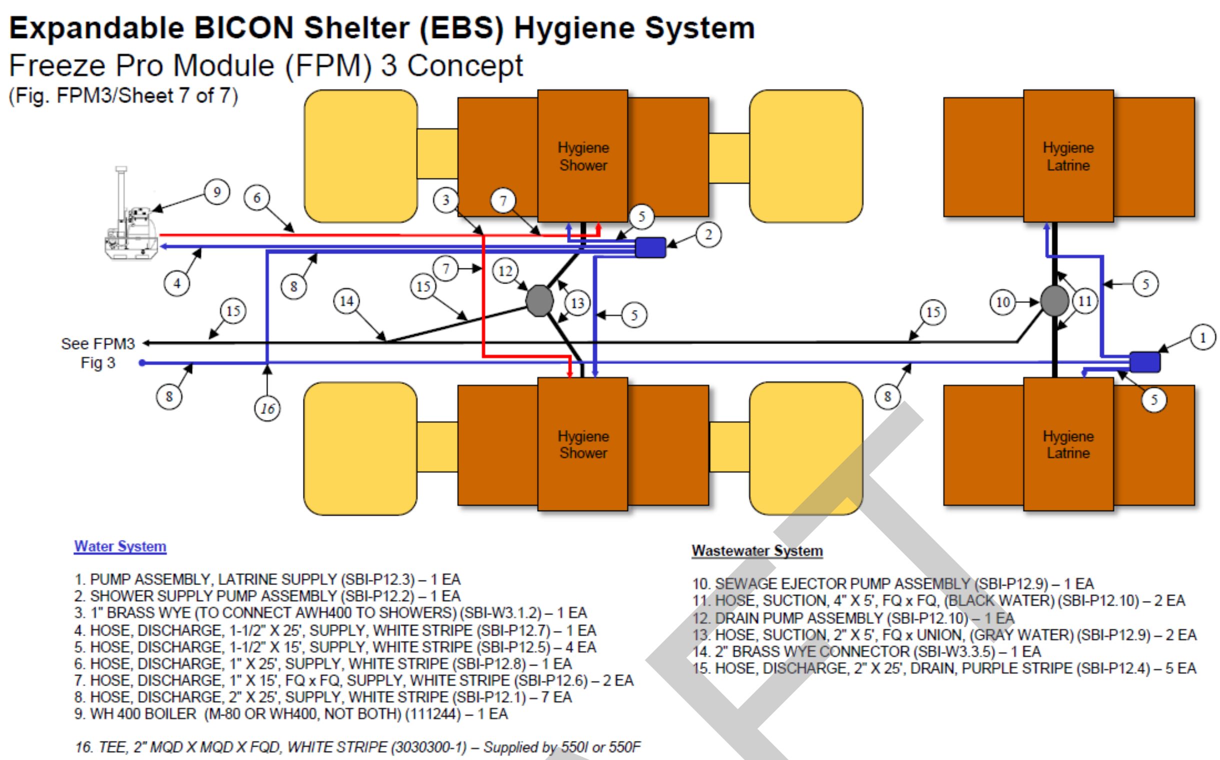


Figure FPM3, 7 of 7: BICON Hygiene Freeze Protection, 550I Freeze Protection Continued.

Table FPM3-Hygiene: 550I Freeze Protection Module, Hygiene Components

Freeze Protection Module 3 (Subsystem: Hygiene System x2)
Freeze Protection Requirements (*Qty to support 1 EA Hygiene)

FP Module	Subsystem	Water Sys Type	Item Category	Hose Dia.	NOMENCLATURE	Figure & Index/Sheet No.	Suggested FP Type	Total FPM Qty	FPM Length (ft)
FPM 3	Hygiene Sys (1)	Water Sys	EQUIP, WATER		PUMP ASSEMBLY, LATRINE SUPPLY	FPM3-1/7	Insulate	1	0
					SHOWER SUPPLY PUMP ASSEMBLY	FPM3-2/7	Insulate	1	0
					WATER HEATER, WH400	FPM3-9/7	Insulate	1	0
			HOSE	1"	HOSE, DISCHARGE, 1" X 25', SUPPLY, WHITE STRIPE	FPM3-6/7	Insulate	1	25
					HOSE, DISCHARGE, 1" X 15', FQ x FQ, SUPPLY, WHITE STRIPE	FPM3-7/7	Insulate	2	30
				1-1/2"	HOSE, DISCHARGE, 1-1/2" X 25', SUPPLY, WHITE STRIPE	FPM3-4/7	Insulate	1	25
					HOSE, DISCHARGE, 1-1/2" X 15', SUPPLY, WHITE STRIPE	FPM3-5/7	Insulate	4	60
				2"	HOSE, DISCHARGE, 2" X 25', SUPPLY, WHITE STRIPE	FPM3-8/7	Insulate	7	175
					Water System Total			17	315
					*Double Qty to support 2 ea Hygiene Systems			*34*	*630*
		Wastewater Sys	EQUIP, WATER		DRAIN PUMP ASSEMBLY	FPM3-12/7	Insulate	1	0
					SEWAGE EJECTOR PUMP ASSEMBLY	FPM3-10/7	Insulate	1	0
			FITTING		2" BRASS WYE CONNECTOR (TO CONNECT LATRINE & SHOWER DRAIN HOSES)	FPM3-14/7	Insulate	1	0
			HOSE	2"	HOSE, SUCTION, 2" X 5', FQ x UNION, SEMI RIGID (GRAY WATER)	FPM3-13/7	Insulate	2	10
					HOSE, DISCHARGE, 2" X 25', DRAIN, PURPLE STRIPE	FPM3-15/7	Insulate	5	125
				4"	HOSE, SUCTION, 4" X 5', FQ x FQ, SEMI RIGID (BLACK WATER)	FPM3-11/7	Insulate	2	10
					Wastewater System Total			12	145
					*Double Qty to support 2 ea Hygiene Systems			24	290
					*Total FPM Qty to support 2 ea Hygiene Systems			*58*	*775*

3.3.4 Module 4, Water Distribution Follow-On Freeze Protection. The FPM 4 for 550F with integration of Hygiene, and Laundry systems shall provide freeze protection through the identified component requirements in accordance with section 3.4. All electrical components required to support freeze protection of the BEAR water system components shall be provided with the respective modules and support interfacing with the standard BEAR electrical distribution system as described in 3.4.2.

3.3.4.1 Freeze Protection for 550F. The FPM 4 shall utilize freeze protection components consistent with FPM 3 without the requirement to provide protection for the dual pump station, dual pump lift station, 25K gallon waste tank system, and Kitchen system.

FPM 4 shall provide freeze protection for the potable water distribution portion of the 550F that includes 3 each - 20K pillow type water bladders with accessory hoses and fittings, 1,000 ft. of 3-inch water hoses, 400 ft. of 4-inch hoses, 400 ft. of 2-inch hoses, and a 125GPM diesel pump. FPM 4 shall include a 32ft. X 20ft. heated and insulated shelter to house 3 each – 3K pillow type water bladders with bladder fill controller assemblies that provide water supply to the Hygiene and Laundry systems.

FPM 4 shall provide freeze protection for the wastewater collection portion of the 550F that includes 1,000 ft. of 3 to 4-inch wastewater hose, 400 ft. of 1 ½ to 2-inch hose, a macerator pump lift station.

Requirements for this module are described in Figures FPM4, sheets 1 - 6, and Tables FPM4-550F Water System, FPM4-550F Wastewater System, FPM4-Laundry, FPM4-Hygiene.

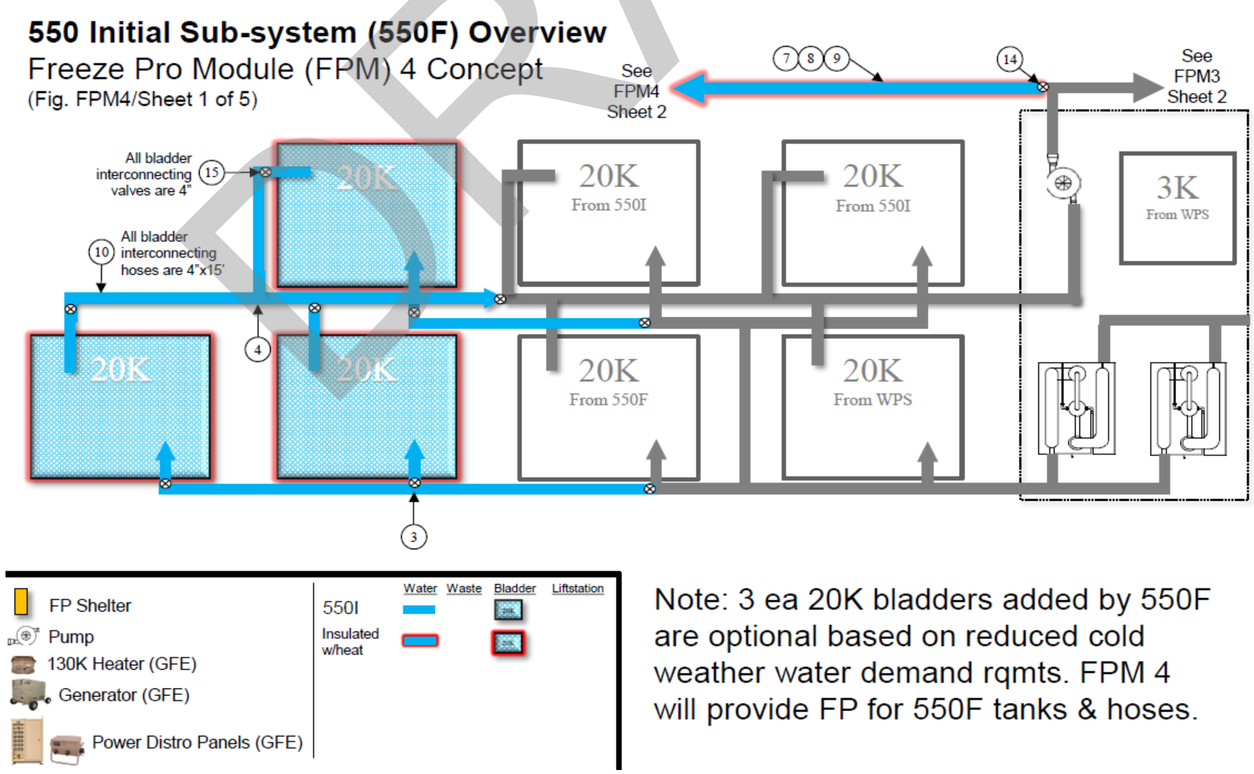


Figure FPM4, 1 of 5: 550F Freeze Protection.

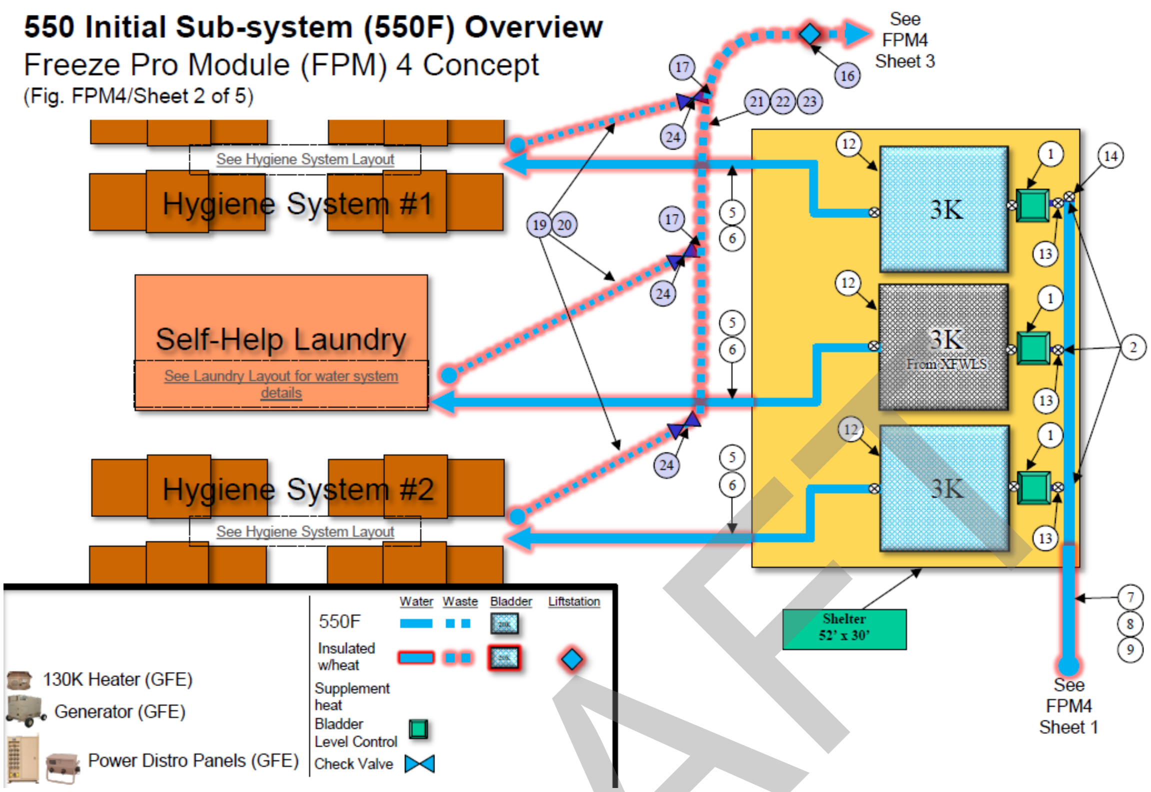


Figure FPM4, 2 of 5: 550F Freeze Protection Continued.

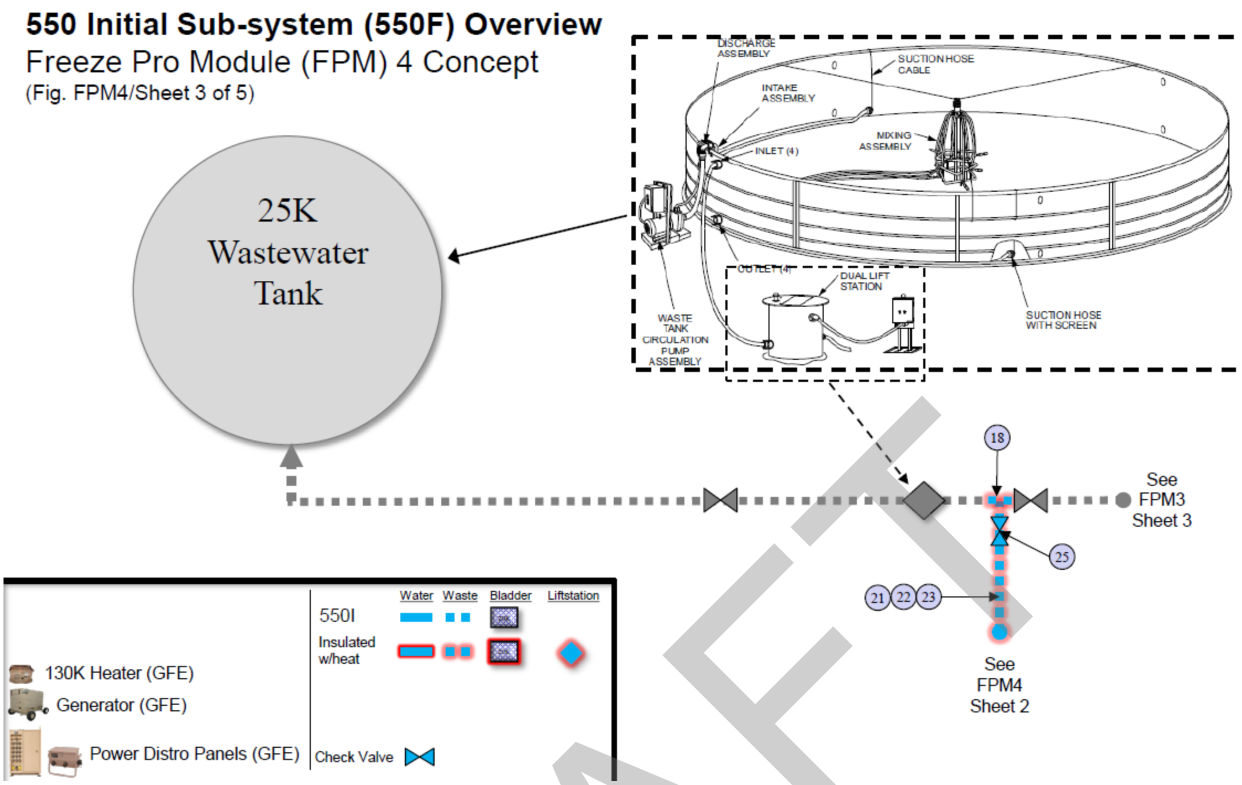


Figure FPM4, 3 of 5: 550F Freeze Protection Continued.

Table FPM4-550F Water System: 550F Freeze Protection Module, Water System Components.

Freeze Protection Module 4 (Subsystem: 550F)

Freeze Protection Requirements

FP Module	Subsystem Name	Water Sys Type	Item Category	Hose Dia.	NOMENCLATURE	Suggested FP Type	Figure & Index/Sheet No.	Total FPM Qty	*FPM Length (ft)
FPM 4	**550F**	*Water Sys*	**EQUIP, WATER**		**CONTROL, BLADDER WATER LEVEL W/25' EX CORD**	Shelter	FPM4-1/2	3	0
			FITTING		**TEE, 3" FQD X 3" MQD X 2" MQD, WHITE STRIP**	Shelter	FPM4-2/2	4	0
					TEE, 4" FQD X MQD X MQD, WHITE STRIPE	Insulate	FPM4-3/1	4	0
					TEE, 4" FQD X MQD X FQD, WHITE STRIPE	Insulate	FPM4-4/1	4	0
			HOSE	2"	**HOSE, SUCTION, 2" X 25', WHITE STRIPE**	Insulate	FPM4-5/2	6	150
					HOSE, SUCTION, 2" X 50', WHITE STRIPE	Insulate	FPM4-6/2	1	50
				3"	**HOSE, DISCHARGE, 3" X 100', WHITE STRIPE**	Insulate	FPM4-7/1,2	7	700
					HOSE, DISCHARGE, 3" X 25', WHITE STRIPE	Insulate	FPM4-8/1,2	4	100
					HOSE, DISCHARGE, 3" X 50', WHITE STRIPE	Insulate	FPM4-9/1,2	4	200
				4"	**HOSE, SUCTION, 4" X 15', WHITE STRIPE**	Insulate	FPM4-10/1	15	225
			TANK		**BLADDER, 20K GAL, WHITE STRIPE**	Insulate	FPM4-11/1	3	0
					BLADDER, 3K GAL, WHITE STRIPE	Shelter	FPM4-12/2	2	0
			VALVE		**VALVE, BALL, 2", WHITE STRIPE**	Insulate	FPM4-13/2	4	0
					VALVE, BALL, 3", WHITE STRIPE	Insulate	FPM4-14/1,2	4	0
					VALVE, BALL, 4", WHITE STRIPE	Insulate	FPM4-15/1	8	0
					Water Sys Total			*73*	*1425*

Table FPM4-550F Wastewater System: 550F Freeze Protection Module, Wastewater System Components.

Freeze Protection Module 4 (Subsystem: 550F)

Freeze Protection Requirements

FP Module	Subsystem Name	Water Sys Type	Item Category	Hose Dia.	NOMENCLATURE	Suggested FP Type	Figure & Index/Sheet No.	Total FPM Qty	*FPM Length (ft)
FPM 4	**550F**	*Wastewater Sys*	**EQUIP, WATER**		**MACERATOR PUMP SEWAGE LIFT STATION**	Insulate	FPM4-16/2	1	0
			FITTING		**WYE, 3" FQD X 3" MQD X 2" FQD, PURPLE STRIPE**	Insulate	FPM4-17/2	4	0
					TEE, 3" FQD X FQD X MQD, PURPLE STRIPE	Insulate	FPM4-18/3	1	0
			HOSE	2"	**HOSE, DISCHARGE, 2" X 25', PURPLE STRIPE**	Insulate	FPM4-19/2	2	50
					HOSE, DISCHARGE, 2" X 50', PURPLE STRIPE	Insulate	FPM4-20/2	5	250
				3"	**HOSE, DISCHARGE, 3" X 100', PURPLE STRIPE**	Insulate	FPM4-21/2,3	5	500
					HOSE, DISCHARGE, 3" X 25', PURPLE STRIPE	Insulate	FPM4-22/2,3	5	125
					HOSE, DISCHARGE, 3" X 50', PURPLE STRIPE	Insulate	FPM4-23/2,3	4	200
			VALVE		**VALVE, CHECK, 2", PURPLE STRIPE**	Insulate	FPM4-24/2	4	0
					VALVE, CHECK, 3", PURPLE STRIPE	Insulate	FPM4-25/3	3	0
					Wastewater Sys Total			*34*	*1125*
FPM 4 Total								**107**	**2550**

3.3.4.3 Freeze Protection for Laundry. Freeze protection requirements for the Laundry System can be found in 3.3.3.3. Requirements for this module are described in Figures FPM4, and Table FPM4-Laundry.

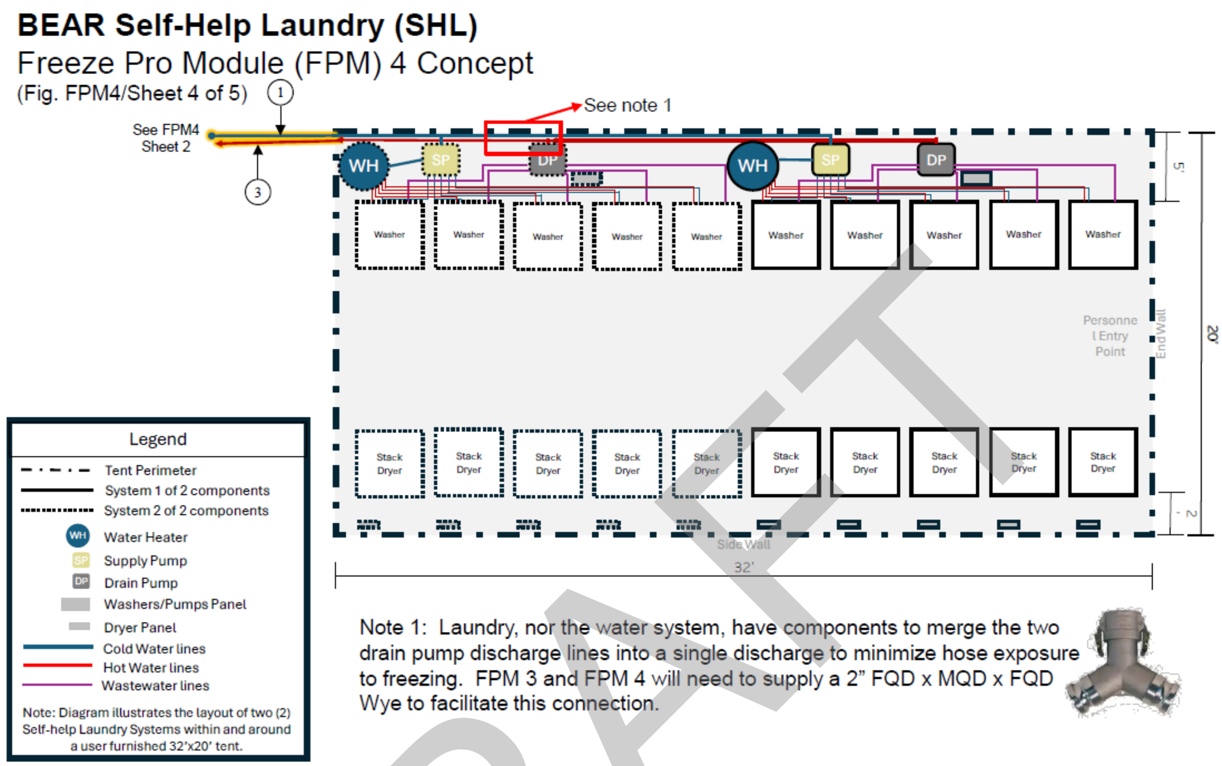


Figure FPM4, 4 of 6: Self-Help Laundry Freeze Protection, 550F Freeze Protection Continued.

Table FPM4-Laundry: 550F Freeze Protection Module, Self-Help Laundry Components

Freeze Protection Module 4 (Subsystem: Laundry)
Freeze Protection Requirements

FP Module	Subsystem Name	Water Sys Type	Item Category	Hose Dia.	NOMENCLATURE	Figure & Index/Sheet No.	Suggested FP Type	Total FPM Qty	*FPM Length (ft)
FPM 4	**Self-Help Laundry**	Water Sys	**HOSE**	1-1/2"	**HOSE, SUCTION, 1-1/2" x 25'**	FPM4-1/4	Insulate	2	50
			TANK		**BLADDER, 3K GAL, WHITE STRIPE**	FPM4-12/2	Shelter	1	0
					Water Sys Total			3	50
		Wastewater Sys	**HOSE**	2"	**HOSE, DISCHARGE, 2" x 25'**	FPM4-3/4	Insulate	2	50
					Wastewater Sys Total			2	50

3.3.4.2 Freeze Protection for Hygiene. Freeze protection requirements for the Hygiene System can be found in 3.3.3.4. Requirements for this module are described in Figures FPM4, and Table FPM4-Hygiene.

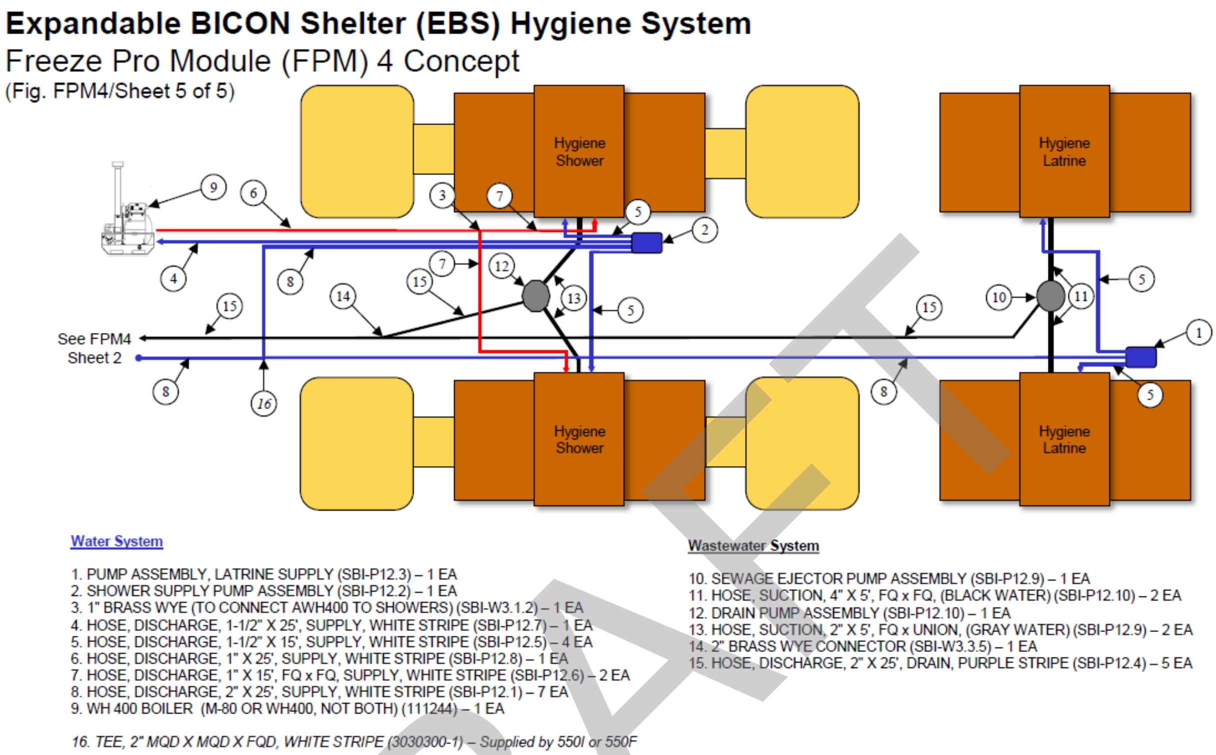


Figure FPM4, 6 of 6: BICON Hygiene Freeze Protection, 550F Freeze Protection Continued.

Table FPM4-Hygiene: 550F Freeze Protection Module, Hygiene Components

Freeze Protection Module 4 (Subsystem: Hygiene System x2)
Freeze Protection Requirements (*Qty to support 1 EA Hygiene)

FP Module	Subsystem	Water Sys Type	Item Category	Hose Dia.	NOMENCLATURE	Figure & Index/Sheet No.	Suggested FP Type	Total FPM Qty	FPM Length (ft)
FPM 3	Hygiene Sys (1)	Water Sys	EQUIP, WATER		PUMP ASSEMBLY, LATRINE SUPPLY	FPM4-1/5	Insulate	1	0
					SHOWER SUPPLY PUMP ASSEMBLY	FPM4-2/5	Insulate	1	0
					WATER HEATER, WH400	FPM4-9/5	Insulate	1	0
			HOSE	1"	HOSE, DISCHARGE, 1" X 25', SUPPLY, WHITE STRIPE	FPM4-6/5	Insulate	1	25
					HOSE, DISCHARGE, 1" X 15', FQ x FQ, SUPPLY, WHITE STRIPE	FPM4-7/5	Insulate	2	30
				1-1/2"	HOSE, DISCHARGE, 1-1/2" X 25', SUPPLY, WHITE STRIPE	FPM4-4/5	Insulate	1	25
					HOSE, DISCHARGE, 1-1/2" X 15', SUPPLY, WHITE STRIPE	FPM4-5/5	Insulate	4	60
				2"	HOSE, DISCHARGE, 2" X 25', SUPPLY, WHITE STRIPE	FPM4-8/5	Insulate	7	175
					Water System Total			17	315
					*Double Qty to support 2 ea Hygiene Systems			*34*	*630*
		Wastewater Sys	EQUIP, WATER		DRAIN PUMP ASSEMBLY	FPM4-12/5	Insulate	1	0
					SEWAGE EJECTOR PUMP ASSEMBLY	FPM4-10/5	Insulate	1	0
			FITTING		2" BRASS WYE CONNECTOR (TO CONNECT LATRINE & SHOWER DRAIN HOSES)	FPM4-14/5	Insulate	1	0
			HOSE	2"	HOSE, SUCTION, 2" X 5', FQ x UNION, SEMI RIGID (GRAY WATER)	FPM4-13/5	Insulate	2	10
					HOSE, DISCHARGE, 2" X 25', DRAIN, PURPLE STRIPE	FPM4-15/5	Insulate	5	125
				4"	HOSE, SUCTION, 4" X 5', FQ x FQ, SEMI RIGID (BLACK WATER)	FPM4-11/5	Insulate	2	10
					Wastewater System Total			12	145
					*Double Qty to support 2 ea Hygiene Systems			24	290
					*Total FPM Qty to support 2 ea Hygiene Systems			*58*	*775*

3.3.5 Module 5, Industrial Operations/Flightline Water Extension Freeze Protection. FPM 5 for industrial operations shall provide protection for all components through a combination of components or methods described in section 3.4. All electrical components required to support freeze protection of the BEAR water system components shall be provided with the respective modules and support interfacing with the standard BEAR electrical distribution system as described in 3.4.2.

3.3.5.1 Freeze Protection for Industrial Ops/Flightline Extension. FPM 5 shall provide freeze protection for two 3,000-gallon pillow type bladders with bladder fill controllers, one 35GPM electric water pump, and 1,000 ft. of 2 – 3 in. diameter hose. Requirements for this module are described in Figure FPM5, and Table FPM5.

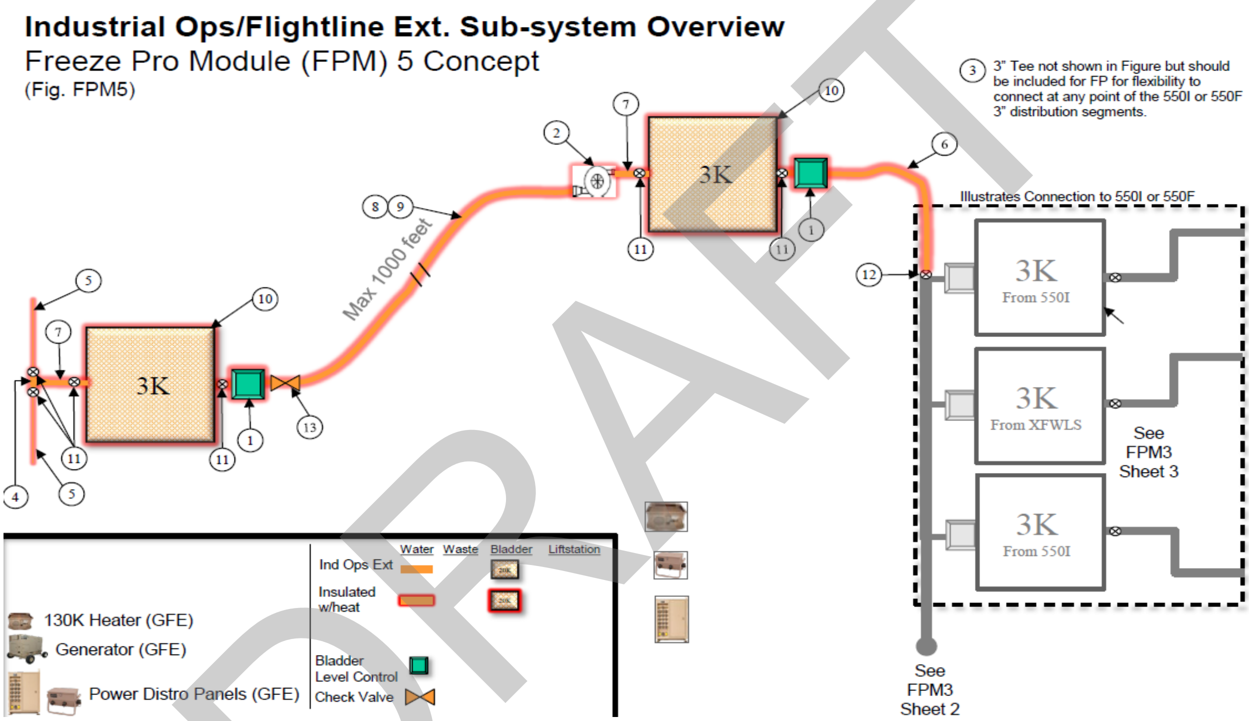


Figure FPM5: Industrial Operations Freeze Protection

Table FPM5: Industrial Operations Freeze Protection Module, Water System Components.

Freeze Protection Module 5 (Subsystem: Industrial Ops/Flightline Ext)
Freeze Protection Requirements

FP Module	Subsystem Name	Water Sys Type	Item Category	Hose Dia.	NOMENCLATURE	Figure & Index/Sheet No.	Suggested FP Type	Total FPM Qty	*FPM Length (ft)
FPM 5	Industrial Ops Ext	Water Sys	EQUIP, WATER		CONTROL, BLADDER WATER LEVEL W/25' EX CORD	FPM5-1	Insulate	2	0
					PUMP ASSY, ELECTRIC, 2" FQD X MQD, WHITE STRIPE	FPM5-2	Insulate	1	0
			FITTING		TEE, 3" FQD X 3" MQD X 2" MQD, WHITE STRIPE	FPM5-3	Insulate	1	0
					TEE, 2" MQD X MQD X FQD, WHITE STRIPE	FPM5-4	Insulate	1	0
			HOSE	1"	HOSE, DISCHARGE, 1" X 25', WHITE STRIPE	FPM5-5	Insulate	2	50
				2"	HOSE, DISCHARGE, 2" X 50', WHITE STRIPE	FPM5-6	Insulate	1	50
					HOSE, SUCTION, 2" X 25', WHITE STRIPE	FPM5-7	Insulate	2	50
				3"	HOSE, DISCHARGE, 3" X 100', WHITE STRIPE	FPM5-8	Insulate	9	900
					HOSE, DISCHARGE, 3" X 50', WHITE STRIPE	FPM5-9	Insulate	2	100
			TANK		BLADDER, 3K GAL, WHITE STRIPE	FPM5-10	Insulate	2	0
			VALVE		VALVE, BALL, 2", WHITE STRIPE	FPM5-11	Insulate	6	0
					VALVE, BALL, 3", WHITE STRIPE	FPM5-12	Insulate	1	0
					VALVE, CHECK, 3", WHITE STRIPE	FPM5-13	Insulate	1	0
					Water Sys Total			30	1150

3.3.7 Module 6, Expansion Kit. FPM 6 shall universally extend freeze protection capabilities of the previously described modules. FPM 6 shall consist of an additional 1,000 feet of heated and insulated hose/pipe wrap, additional 20,000-gallon and 3,000-gallon bladder freeze protection equipment. The expansion kit shall consist of components previously developed to support freeze protection of Modules 1 through 5.

FPM 6 shall provide protection through a combination of components or methods described in section 3.4. All electrical components required to support freeze protection of the BEAR water system components shall be provided with the respective modules and support interfacing with the standard BEAR electrical distribution system as described in 3.4.2.

3.4 Performance. The BEAR Water Freeze Protection Modules shall provide freeze protection for the BEAR water subsystem components in an operating temperature range from 32°F to -25°F (objective goal of -65°F).

3.4.1 BEAR Water Freeze Protection Installation. The Water Freeze Protection Modules shall be designed to support quick and easy installation of all freeze protection components. The Modules shall be designed to support installation without major interruption to water system operation. Installation of freeze protection material onto any 3-inch diameter hose, 100-feet in length, shall not exceed 5 minutes with two trained personnel, with gloved hands.

3.4.2 BEAR Water Freeze Protection Electrical Requirements. The Water Freeze protection modules' electrical components shall be compatible and interface with the existing BEAR electric power system equipment to receive electrical power as required for module operation. Electrical power for each module shall be derived from Government Furnished Equipment (GFE) Mobile Electric Power (MEP) generators (see Table 1) to provide tactical power at the point of use locations and secondarily be capable of interfacing with the BEAR Electrical Distribution system as described in AF TTP 3-32.34V5. GFE components listed in Table 2 shall be utilized in the design of module power system and identified and listed within the respective module's baseline index as GFE. BEAR power generation and distribution is a nominal 120/208 VAC single or three phase 60 Hertz, additional details may be provided upon request. Any equipment required that is not listed in Table 2 shall be furnished with the respective module.

The freeze protection solution shall maximize energy efficient solutions. The contractor shall provide total power requirements for all freeze protection modules individually and a total requirement for FPM 1 through 6.

No single electrical component, device or single fed system shall draw more than 60 amps. All exposed electrical components shall be weatherproof and shall be UL listed for their intended use wherever possible. All electrical installations shall comply with NFPA 70. Military specification connectors shall be utilized for all electrical components; however, some commercial, three-prong, 120V connectors may be used with procuring activity's approval.

MOBILE ELECTRIC POWER (MEP) GENERATORS, ADVANCED MEDIUM MOBILE POWER SOURCES (AMMPS) FAMILY				1 PHASE AMPERAGE		3 PHASE AMPERAGE	
Model	*NSN*	*kW*	*Remarks*	***120V***	***120V/240V***	***120V/208V***	***240V/416V***
MEP-1030	6115-01-561-7329	5	120 VAC 1 ph, 120/240 VAC 1 ph, 120/208 VAC 3 ph @ 50/60 Hz	52	26	17	N/A
MEP-1030A	6115-01-679-2558	5	120 VAC 1 ph, 120/240 VAC 1 ph, 120/208 VAC 3 ph @ 50/60 Hz	52	26	17	N/A
MEP-1040	6115-01-561-7455	10	120 VAC 1 ph, 120/240 VAC 1 ph, 120/208 VAC 3 ph @ 50/60 Hz	104	52	35	N/A
MEP-1040A	6115-01-679-5145	10	120 VAC 1 ph, 120/240 VAC 1 ph, 120/208 VAC 3 ph @ 50/60 Hz	104	52	35	N/A
MEP-1050	6115-01-561-7634	15	120/208 VAC 3 ph, 240/416 VAC 3 ph @ 50/60 Hz	N/A	N/A	52	26
MEP-1050A	6115-01-679-5596	15	120/208 VAC 3 ph, 240/416 VAC 3 ph @ 50/60 Hz	N/A	N/A	52	26
MEP-1060	6115-01-561-7718	30	120/208 VAC 3 ph, 240/416 VAC 3 ph @ 50/60 Hz	N/A	N/A	104	52
MEP-1060A	6115-01-679-5442	30	120/208 VAC 3 ph, 240/416 VAC 3 ph @ 50/60 Hz	N/A	N/A	104	52
MEP-1070	6115-01-561-7788	60	120/208 VAC 3 ph, 240/416 VAC 3 ph @ 50/60 Hz	N/A	N/A	208	104
MEP-1070A	6115-01-679-4747	60	120/208 VAC 3 ph, 240/416 VAC 3 ph @ 50/60 Hz	N/A	N/A	208	104

Table 1

BEAR Electrical Distribution Equipment

Item	***NSN***	***Input Connection***	***Output Connection***
25KW Power Distribution Panel (PDP)	6110-01-244-3209 (P/N 8642307)	120/208V, 3PH, 60A, MS90558C32412P (1 EA)	120/208V, 3PH, 60A, MS90555C32412S (1 EA) 120V, 1PH, 15A , MS25199-1 (1 EA) 120V, 1PH, 20A, 8642377-1, NEMA L5-20R (4 EA)
60KW Power Distribution Panel (PDP)	6110-01-237-4637 (P/N 8542414)	120/208V, 3PH, 200A, MS90558C52412P (1EA)	120/208V, 3PH, 60A, MS90555C32412S (4 EA)
Secondary Distribution Center (SDC)	6110-01-168-8077 (P/N 8335844)	4160V Load Breaks (3 Phase) or 120/208V, 3PH, 200A, MS90558C52412P (1EA)	120/208V, 3PH, 60A, MS90555C32412S (16 EA) 120V, 1PH, 15A , MS25199-1 (2 EA)
Cable, 200A, 25 ft	6150-00-178-9159	120/208V, 3PH, 200A, MS90556C52412P (1 EA)	120/208V, 3PH, 60A, MS90557C52412S (1 EA)
Cable, 60A, 50 ft	6150-00-255-8313	120/208V, 3PH, 60A, MS90556C32412P (1 EA)	120/208V, 3PH, 60A, MS90557C32412S (1 EA)
Cable, 60A, 100 ft	6150-00-255-8332	120/208V, 3PH, 60A, MS90556C32412P (1 EA)	120/208V, 3PH, 60A, MS90557C32412S (1 EA)

Table 2

3.4.3 BEAR Water System Configuration and Approved Layout Deviations/Alternate Technical Approaches. The BEAR Freeze Protection Modules may propose alternate approved layouts and/or approved alternate technical approaches that increase effectiveness of protection or accommodate recirculation. The baseline configuration of the BEAR water system shall not be modified, any deviations that require additional piping, valves, or other components shall be provided by the freeze protection kits/modules. Alternate layouts may include consolidation of components across FPMs into collective protection such as sheltering, to accommodate efficiency and or more robust protection.

3.4.4 BEAR Freeze Protection Equipment. The BEAR Freeze Protection Modules may contain equipment including but not limited to insulated valve boxes, electrical and fuel-powered pump insulation covers, bladder covers, immersion heaters, recirculation devices, insulated hose wrap, heat tape, shelters, etc. The freeze protection system shall utilize universal or common components across freeze protection kits/modules. Unique items shall be limited as much as possible.

3.4.4.1 Hose and Fittings Freeze Protection. The BEAR Water Subsystems contain various size hoses and fittings to support distribution of potable and combined grey and black wastewater by-products. The BEAR Freeze protection modules shall provide protection for hoses and fittings in ambient temperatures ranging from 32°F to -25°F. The hose protection solution shall allow for easy install and removal with gloved hands, without breaking hose connections. The hose protection solution shall be made of materials that are UV resistant outer layer and resist splitting or cracking in arctic climates. The hose protection solution shall be designed to be installed on existing water/waste hoses and fasteners together via hook and loop strips along its entire length, or other expedient installation method. The hose protection solution shall be standardized to support various hose sizes to the maximum extent, i.e. 3 in. and 6 in. hoses utilize the same Part Number (P/N) for hose wrap insulation. Any proposed solution shall be approved by the procuring government activity. Wrap that identifies the requirement for heat trace shall conform with 3.4.4.2.

3.4.4.2 Heat Tape/ Heat Trace. The BEAR Freeze Protection Modules may utilize self-regulating, electrical heat tape or heat trace to increase system effectiveness. The heat tape/trace shall be terminated with male/female weatherproof plugs/receptacles so the units can be daisy chained to accommodate longer hose/pipe runs. Electrical power draw of the self-regulating heat tape shall not exceed 9 watts per foot. Electrical distribution system for heat tape/trace shall not exceed 120/208 volts, 60 amps per connection from BEAR electrical Power Distribution Panel (PDP). Heat trace may be integrated into insulated wraps or covers.

3.4.4.3 Water Bladder and Waste Tank Freeze Protection. The BEAR Water Subsystems contain 3K and 20K gallon pillow type bladders and a 25K gallon open top waste tank. The BEAR freeze protection modules shall provide insulation material or heated shelters to prevent contents of bladders or waste tanks from freezing and minimize energy loss. All covers shall allow easy access for normal operations, and water sampling. Usage of heat additive systems may be utilized to increase system effectiveness. Heat additive systems include but are not limited to heat pads, water boilers/heaters, heat trace, etc.

3.4.4.4 Electrical and Diesel Driven Pumps Freeze Protection. The BEAR Water Subsystems contain various size pumps to support water distribution or waste ejection. The BEAR freeze protection modules shall provide form fitting heated insulated blankets or boxes designed for each respective pump. Pumps may also be protected by use of a heated shelter listed and in accordance with 3.4.4.6.

3.4.4.5 Various Valve Freeze Protection. The BEAR Water Subsystems contain various valves to support water distribution or waste ejection. The BEAR freeze protection modules shall provide insulated and heated valve boxes for each valve not protected by a heated shelter. The design of the valve box shall allow easy access, and operation of valves as required.

3.4.4.6 Shelters for Freeze Protection. To protect collective components from freezing, BEAR Water Subsystems' operational layouts require heated shelters. Government-qualified shelters shall be used as the primary solution. However, alternative shelters may be recommended for approval if they offer superior performance, such as greater efficiency or improved snow load capacity.-All shelters, whether soft or hard-sided, must be equipped with a cold-weather insulation package. Each shelter must be evaluated for its performance in the intended operating environment; it is recommended that this evaluation references the testing standard TOP 10-2-175. The key performance requirement is the ability to maintain an internal temperature above 32°F with an external ambient temperature as low as –25°F. Shelters will be heated by the government-furnished (GFE) 130K BTU multi-fuel burning heater (NSN 4520-01-521-2076), and any alternative heating source such as inferred, radiant, alternative forced air, etc. must be approved. Shelters can be combined or complexed to fit the required footprint and shall be provided as an integral component of the freeze protection module.

BEAR qualified shelter systems:

Small Shelter System, NSN 8340-01-469-1876 (tan), Dimensions: L 32 ft. x W 20 ft. x H 10ft.
Medium Shelter System, NSN 5419-01-465-3019 (tan), Dimensions: 52 ft. x 30 ft. x 15 ft.
4K Dome Shelter System, NSN 5410-01-455-2004, Dimensions: 89 ft. x 70 ft. x 25 ft.
8K Dome Shelter System, NSN 5410-01-494-5130, Dimensions: 116 ft. x 69 ft. x 25 ft.
Large Area Maintenance Shelter, NSN 5410-01-533-7395, Dimensions: 129 ft. x 75 ft. x 31 ft.

3.4.4.7 Heat Pad. The BEAR Water Freeze Protection Modules may utilize self-regulating, electrical heating pads to increase system effectiveness. Heat pads may be used to provide freeze protection to various BEAR water system components including bladders, waste tanks, and pumps. Installation and usage instructions must be provided. Electrical distribution system for heat pads shall not exceed 120/208 volts, 60 amps per panel.

3.4.4.8 Water Boilers/Heaters. The BEAR Water Freeze Protection Modules may utilize the AWH400 (4520-01-566-6669) or M-80 (4520-01-162-0385) to increase system effectiveness. Water Boilers/Heaters may be used to increase overall temperature of water in the BEAR water system. When used in shelters, exhaust adapters shall be provided to redirect exhaust outside of shelter.

3.4.4.9 Immersion Heaters. The BEAR Freeze Protection Modules may utilize immersion style heaters to improve freeze protection capabilities. Each immersion style heater shall be self-regulating, and properly sized for the bladder or storage tank, and clearly identified to prevent misuse.

3.4.4.10 Flow Control Devices, Pumps and/or Equipment. The BEAR Freeze Protection Modules may utilize water recirculation to increase effectiveness. The flow control devices and equipment shall be compatible with the current BEAR systems. The flow control devices, pumps and/or equipment shall be capable of operation without freezing in the desired operating environment. The flow control devices, pumps and/or equipment shall be automatically actuated based on ambient air temperature set point.

3.5 Environmental conditions.

3.5.1 Operating temperature range. The BEAR Freeze Protection Modules shall prevent the BEAR Water subsystems from freezing in ambient temperatures ranging from 32°F to -25°F (objective goal of -65°F).

3.5.2 Storage temperature range. The BEAR Freeze Protection Modules shall be capable of being stored in ambient temperatures ranging from -65° F to 160° F.

3.5.3 Precipitation.

3.5.3.1 Rain. The BEAR Freeze Protection Modules shall be capable of storage and operation during rainfall of 5-inches per hour for three consecutive hours and 10-inches per hour for 10 consecutive minutes, with winds of up to 35 knots; and with 6-inches of rain per hour impinging on the BEAR Freeze Protection Modules at angles from vertical to 45° for 30 consecutive minutes.

3.5.3.2 Snow. The BEAR Freeze Protection Modules shall be capable of long-term storage and operation during accretion of wet snow up to 2-inches per hour for at least 12 hours on exposed horizontal surfaces.

3.5.3.3 Ice. The BEAR Freeze Protection Modules shall be capable of long-term storage and operation with ice accretion up to 1.5-inches on exposed horizontal surfaces.

3.5.4 Solar radiation. The BEAR Freeze Protection Modules shall not be adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments and high altitudes.

3.5.5 Fungus. All materials used in the BEAR Freeze Protection Modules shall be fungus resistant or shall be suitably treated to resist fungus. Materials treated for fungus resistance shall retain their original electronic and physical properties, shall not present toxic hazards, and treatment shall last for the entire service life of the part. The BEAR Freeze Protection Modules shall be suitable for operation and storage in conditions encountered in a tropical environment.

3.5.6 Salt fog. The BEAR Freeze Protection Modules shall be capable of sustained use and storage in high temperature, high humidity, salt laden, sea-coast environments without damage or deterioration of performance. The BEAR Freeze Protection Modules shall also be capable of storage and operation in part-time (5%) exposure to a five percent salt solution, with fallout rate of 0.10 oz (3 ml) of per hour per each 12.40 sq. inch (80 sq. cm) of horizontal collecting area without damage or deterioration of performance.

3.5.7 Sand and dust. The BEAR Freeze Protection Modules shall be capable of sustained use and storage with exposure to 0.062 grams/cubic foot wind-blown sand (particle size 150-1000 microns) or dust (particle size less than 150 microns) at wind speeds up to 35 knots without damage or deterioration of performance.

3.5.8 Humidity. The BEAR Freeze Protection Modules shall be capable of operation and storage from 5 to 95% (±5%) relative humidity, including conditions where condensation takes place in the form of water and/or frost.

3.6 Weight and dimensions. The size and weight of the BEAR Water Freeze Protection Subsystems shall be as small as practicable consistent with the requirements specified herein. Maximum size and weight shall not exceed TABLE I requirements for Surface transport configurations and TABLE II requirements for Air Transport configurations. The packaging container weight shall be included in the total weight of each subsystem UTC.

TABLE I. Maximum weight and dimensions for Surface Configurations.

TRICON Weight	14, 900 pounds.
BICON Weight	25,000 pounds.
20’ Container Weight	25,000 pounds.

TABLE II. Maximum weight and dimensions for Air Configurations.

463L/ISU-90 Weight	10,000 pounds.

3.7 Packaging, Marking and Transportability. The BEAR Water Freeze Protection Modules shall be designed to properly store and ship in ISU-90 containers for air transport OR TRICONS, BICONS and/or 20’ ISO containers for surface transport configurations. Packaging instructions for both container types shall be provided. Specific packaging configuration mode will be identified within delivery order contract terms.

The BEAR Water Freeze Protection Modules shall be air transportable mode on aircraft; or surface mode by truck, trailer, railcar, and ship and all shall be capable of withstanding the mechanical shock, vibration, temperature, humidity, and pressure characteristics of these modes of transportation.

3.7.1 Air Transport Mode. All Unit Type Codes (UTC) shall be configured for air transport in accordance with MIL-STD-1791. Air configured packaging for each Freeze Protection Module will fit 463L Pallet/ISU-90 containers.

ISU-90, NSN 8145-01-512-1201, in accordance with MIL-STD-1791

3.7.2 Surface Transport Mode. The Freeze Protection Modules shall be packed for multimodal transportation. For surface transportation configuration the sub-systems shall be packed in TRICONS, BICONs, or combination of both type containers.

TRICON, Type 5, NSN 8150-01-574-1682, in accordance with MIL-PRF-32402
BICON, Type 2, NSN 8150-01-586-4979, in accordance with MIL-PRF-32410
20’ ISO Container, NSN 8150-01-501-2741, in accordance with A-A 52032B

3.7.3 Preservation, packing and marking. The Freeze Protection Modules interior packages and shipping containers shall be preserved, packed and marked for shipping in accordance with the AFMC Form 158. The shipping containers shall be packed in accordance with the BEAR WSO Packing Plans and shall include watershed shrouding per the AFMC Form 158.

3.8 Maintainability and service life.

3.8.1 Maintainability. The BEAR Freeze Protection Modules shall be designed for maintainability in accordance with 5.9 through 5.9.22.9 of MIL-STD-1472; forces shall not exceed those specified for both males and females while wearing arctic clothing.

The BEAR Freeze Protection Modules shall be designed to use common tools to the maximum extent possible, if special tools are required, they shall be included in the respective subsystem/module components.

3.8.1.1 Preventive maintenance. The recommended routine preventive maintenance interval (PMI) shall be at least 500 operating hours. Preventive maintenance tasks, other than tasks required daily or before operation, shall not require more than 1 man-hours to complete while wearing arctic clothing. This shall be calculated by:

$$PMMH_T = \sum_{i=1}^{N} \frac{PMMH_i}{i}$$

where:

$PMMH_T$	is the total preventive maintenance man-hours,
i	is a multiple of the routine PMI (that is, i=1 for the routine PMI, i=2 for every other routine PMI, etc.), and
$PMMH_i$	is the preventive maintenance man-hours at the ith PMI.

Routine servicing and inspections shall not require hand tools.

3.8.2 Service life. The BEAR Freeze Protection Modules shall be designed for a minimum service life of 10 years without a need for major overhaul or repair. Service life does not apply to wear items such as filters, electric motor brushes, fuses, hoses, cables, etc. The system design shall be suitable for a minimum of 25 years outside storage.

3.9 Human systems integration (HSI). The BEAR Freeze Protection Modules shall be designed in accordance with MIL-STD-1472 for ease of operation, inspection, and maintenance, including the use of arctic mittens.

3.10 Removed.

3.11 Safety.

3.11.1 System safety. The design of the BEAR Freeze Protection Modules shall not contain any system safety mishap risk categories greater than medium as defined in MIL-STD-882.

3.11.2 Hazardous materials. The design shall minimize, and control hazards associated with the inclusion or use of hazardous or toxic materials and the generation of toxic or noxious gases. The BEAR Freeze Protection Modules shall not generate or use Class I or Class II Ozone Depleting Substances (ODS) during operation, maintenance, or disposal. Class I ODS and hazardous materials shall not be used in any system, component, or process. The BEAR Freeze Protection Modules shall not contain or use either hexavalent chromium or cadmium without written approval by the procuring activity. The Environmental Protection Agency maintains an online list of toxic chemicals and hazardous substances at http://www.epa.gov/emergencies/tools.htm. Class I and Class II ODS are defined in 40CFR82.

3.11.3 Component protection. The BEAR Freeze Protection Modules shall be designed so that all electrical/mechanical output components are enclosed, and access is not required during normal operation. Further, the BEAR Freeze Protection Modules shall be designed to safely permit access to internal components that may be required during troubleshooting for failures or other special needs. All space in which work is performed during operation, service, and maintenance shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to eliminate the hazard without impairing the function of the parts. All wires, cables, tubes, and hoses shall be supported and protected to minimize chafing and abrasion and shall be located to provide adequate clearance from moving parts and high operational temperatures. Grommets shall be provided wherever wires, cables, tubes, or hoses pass through bulkheads, partitions, or structural members.

3.12 Protective coatings, finish, and corrosion control.

3.12.1 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during the service life of the BEAR Freeze Protection Modules shall not be used or shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein.
The BEAR Freeze Protection Modules shall be constructed of corrosion resistant materials to the maximum extent possible. Surface treatments, pre-treatments and coatings shall be applied to all components that are not corrosion resistant. Surface treatments include electroplated coatings, thermal sprayed coatings, and hard anodize coatings. Surface pre-treatments include thin film anodize, conversion coatings, zinc phosphate, and other coatings used to enhance subsequent corrosion protective coating adhesion and performance. Corrosion protective coatings include corrosion inhibited epoxy primers and polyurethane topcoats. Corrosion protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used. Exposed surfaces of fasteners, handles, and fittings shall also be primed and painted. Commercial items (see 6.3.1) used as components of the BEAR Freeze Protection Modules (but not the BEAR Freeze Protection Modules itself) may be prepared and coated in accordance with the manufacturer's standard practice, provided it is compatible with the exterior finish color. Some commercial items (see 6.3.1) shall have the protective coatings required herein in accordance with the respective material substrates.

3.12.2 Surface preparation. Components shall be cleaned, degreased, and prepared prior to any scuffing, sanding, blasting, or application of any pretreatment. Surface preparation and pretreatment shall be in accordance with the respective primer and topcoat specifications. All corrosion protection steps, regardless of material, shall be applied before any new oxidation or rusting occurs post-treatment and shall be completed prior to application of any coating used as either primer or topcoat. Surfaces shall be scuffed, sanded, and/or blasted, depending upon material type and thickness, to remove any existing surface impurities such as: primer/paint/topcoat, corrosion, welding/grinding/cutting slag, etc. that may prevent adhesion and protection of further corrosion control measure. Any surface undergoing a media blasting process shall have the blasting pressure, blasting material type, and blasting material grit suitably chosen to prevent warpage or other structural damage. Coatings requiring a 'near-white' ferrous metal finish shall have the surface(s) prepared in accordance with SSPC-SP 10/NACE No. 2. General surface finish and cleanliness shall be in accordance with SSPC-Vis 1.

3.12.2.1 Aluminum pretreatment. Aluminum surfaces used in construction of the BEAR Freeze Protection Modules shall be treated with a suitable conversion coating prior to application of any other corrosion prevention coating. Anodizing or a MIL-DTL-81706 conversion coating process may be utilized, at the discretion of the BEAR Freeze Protection Modules manufacturer. Any anodizing shall be applied in accordance with MIL-A-8625, Type II, Class 1; no further conversion coating is required on anodized aluminum surfaces. Alternatively, aluminum surfaces shall have MIL-DTL-81706, Type II, Class 1A, Form I, Method C chemical conversion coating applied in accordance with MIL-DTL-5541, Type II, Class 1A.

3.12.2.2 Ferrous pretreatment. Ferrous surfaces used in construction of the BEAR Freeze Protection Modules shall be treated with a metal etching compound prior to application of any further pretreatment process(es). Ferrous surfaces with iron or zinc phosphate coatings/etches do not require further pretreatment. Ferrous surfaces without iron or zinc phosphate coatings/etches shall be treated with a wash primer in accordance with TT-C-490 to seal out flash rust.

3.12.3 Primer. Raw metal edges, to include fastener and drain holes, shall be coated with primer prior to applying the topcoat. Any surfaces where primer and/or topcoat is not required, or intended to be applied, shall be masked off using materials and processes suitable for use with applied primer and/or topcoat system. A primer color shall be chosen that is compatible with the color of the topcoat used such that scratches in the topcoat do not yield an obtrusive contrast with any exposed primer (tone-down primer in a similar shade/color family as the topcoat).

3.12.3.1 Ferrous surfaces. Ferrous structures and surfaces shall be primed with a non-water reducible zinc rich primer in accordance with MIL-DTL-53022, Type IV, Class L. The primer system shall be allowed to dry and fully cure in accordance with the primer manufacturer's directions prior to applying the topcoat.

3.12.3.2 Aluminum and mixed aluminum and ferrous surfaces. Aluminum and mixed aluminum and ferrous structures and surfaces shall be primed with an epoxy primer in accordance with MIL-DTL-53022, Type IV, Class L.

3.12.4 Topcoat. All Freeze Protection components, except for commercial items (see 6.3.3), plastic, rubber, slide able (brake linkages), and friction surfaces (brake surfaces) shall be top coated with polyurethane in accordance with MIL-PRF-85285, Type IV, Class H. Neither Chemical Agent Resistant Coating (CARC) nor powder coating shall be used. The coating shall be free from runs, sags, orange peel, or other defects. The contractor shall utilize the topcoat material manufacturer's specifications to accomplish application of the coating system in accordance with the manufacturer's directions to the minimum dft (dry-film thickness) required herein.

3.12.5 Finish. All finishing processes utilized in the design and construction of the unit shall result in a durable, non-fading, long lasting finish. Finish warranty shall be equivalent to best commercial warranty offered or a minimum of 5 years, whichever is greater. Warranty is not to include damage caused by common maintenance activities. The exterior finish color of the BEAR Freeze Protection Modules and the inner surfaces of compartments shall be Tan 686A Camo, Color Number 33446, or Green 383 Camo, Color ID 34094, of SAE AMS-STD-595. Specific color requirements will be identified during procurement activity.

3.12.6 Exclusion of water. The design of the BEAR Freeze Protection Modules shall be such as to prevent water leaking into, or being driven into, any part of the BEAR Freeze Protection Modules that would degrade overall system operating performance. All doors, panels, covers, etc., shall be provided with sealing arrangements such that the entry of water is minimized or eliminated when these items are correctly closed. Particular care shall be taken to prevent wetting of equipment and heat and sound proofing materials. Sharp corners and recesses shall be avoided so that moisture and solid matter cannot accumulate to initiate localized attack. Sealed floors with suitable drainage shall be provided for storage compartments and other areas in the BEAR Freeze Protection Modules that could collect and retain water.

3.12.7 Fluid traps and faying surfaces. There shall be no fluid traps on the BEAR Freeze Protection Modules. Faying surfaces of all structural joints, except welded joints, shall be sealed to preclude fluid intrusion.

3.12.8 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

3.12.9 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. All designs shall include considerations for the prevention of water or fluid entrapment and ensure that drain holes are located to effect maximum drainage of accumulated fluids. The number and location of drain holes shall be sufficient to permit drainage of all fluids when the BEAR Freeze Protection Modules is on a 10° longitudinal slope facing both up and down and on a 10° side slope in each direction (right and left side facing up the slope). The minimum size of the drain holes shall be 0.375 inch.

3.12.10 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be in contact with each other. Metal plating or metal spraying of dissimilar base metals to provide electromotively compatible abutting surfaces is acceptable. The use of dissimilar metals when separated by suitable insulating material is permitted, except in systems where bridging of insulation materials by an electrically conductive fluid can occur. Sealants or gel type gasket materials shall be used between faying surfaces and butt joints.

3.13 Markings. All external devices which require an operational or maintenance interface (that is, cautions, warning, etc.) shall be marked in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1-inch high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible but shall not be less than one-half inch high. Markings, Information/Caution shall be Lusterless Black, Color Number 37038 of SAE AMS-STD-595, and Markings, Warning/Danger shall be Lusterless Red, Color Number 31136 of SAE AMS-STD-595.

3.14 Identification and information plates.

3.14.1 Identification plate. The BEAR Freeze Protection Modules and major components shall be provided with a permanently-marked identification data plate constructed of a non-corroding metal permanently affixed to the exterior of the BEAR Freeze Protection Modules in a readily accessible location in accordance with MIL-STD-130. The identification plate shall contain the following information: nomenclature, part number, serial number, date of manufacture, manufacturer's name, Commercial and Government Entity (CAGE), date of warranty expiration, contract number, and National Stock Number (NSN). In addition, the identification plate shall include six (6) blank lines across the bottom for future etching by the USAF. The identification plate shall be mounted at a conspicuous place on the exterior of the BEAR Freeze Protection Modules. The 'CE' marking shall be affixed in accordance with EU requirements on or adjacent to the identification plate.

3.14.2 Unique Identification (UID). Unique Identification (UID) BEAR Freeze Protection Modules and any of its components for which the Government's unit cost is more than $5,000, is serially managed, or the procuring agency determines is mission essential, shall have Unique Identification (UID) (also known as Item Unique Identification (IUID)) information permanently affixed on or near the respective identification plate(s), marked in accordance with MIL-STD-130. UID information shall be included as both a bar code and human readable markings and contain the following information:

a. National Stock Number (NSN).

b. Manufacturer's part number.

c. Serial number.

d. Manufacturer's CAGE.

3.15 Design and construction. The design shall promote cost effective, life-cycle sustainability by addressing considerations such as incorporating open standards while satisfying system performance requirements. It shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. It shall be weatherproof and designed to prevent the intrusion of water, sand, and dust into critical operating components.

3.15.1 REMOVED.

3.15.2 Mechanical design.

3.15.2.1 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for the attachment of trim items or as provided in commercial (see 6.3.1) components. When applicable, fasteners shall be installed with the application of a design fastening torque such that an adequate preload is introduced in the fastener to prevent losing or fatiguing the fasteners. Externally threaded fasteners for use with internally threaded nuts or holes shall be in accordance with SAE J429. Internally threaded nuts shall be in accordance with SAE J995. Tapped threads shall have a minimum thread engagement in accordance with Table III. To the maximum extent possible, fastening designs that utilize bolted connections shall be accomplished using a through-bolt method, and secured with locking nuts or weld-nuts. The use of riv-nuts and tapped threads is discouraged. The use of socket-head, flat, countersunk fasteners shall be minimized. When used, socket-head, flat, countersunk fasteners shall be in accordance with NASM 24667, or ASME B18.3 and ASTM F835, and shall be used with a corresponding nut in lieu of a threaded hole to alleviate potential fastener/hole misalignment and bending stress on the head. When in contact with aluminum materials, steel fasteners shall be coated with a Zinc/Aluminum Corrosion Protective Coating in accordance with ASTM F1136 Grade 3.

TABLE III. Minimum thread engagement.

Material	**Minimum Thread Engagement**
Steel	1.0 times the nominal fastener diameter
Cast iron, brass, or bronze	1.5 times the nominal fastener diameter
Aluminum, zinc, or plastic	2.0 times the nominal fastener diameter

3.15.2.2 Welders and welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings. Cast components, structural steel components with an ultimate tensile strength greater than 100,000 psi, and threaded fasteners (except for weld studs and weld nuts) shall not be welded. Heat treated components shall be in the annealed, normalized condition when welded.

3.15.2.3 Hook and Loop. All Hook and Loop fasteners shall be moisture and UV resistant and capable of withstanding normal working loads without separation. Hook and Loop material shall be fixed to hose wraps, cover, etc. in a fashion that will ensure a permanent bond for the service life period. See A-A-55126C - FASTENER TAPES, HOOK AND LOOP, SYNTHETIC.

3.15.3 Wiring. All wires used in the BEAR Freeze Protection Modules shall be multi-stranded copper conductor, flame retardant, moisture, heat, and oil resistant in accordance with NFPA 70. All wires shall be numbered for easy identification. The marking shall be permanently applied to the wire to withstand all specified operating conditions herein throughout the life expectancy of the end item. Marking shall be applied within 3 inches of each end and at intervals not to exceed 12 inches throughout the wire length. Wires having a total length of less than 18 inches shall be marked within 3 inches of both ends and once in the (approximate) middle of the wire. The wire-numbering scheme shall be consistent with that currently being employed by industry standards. Wires energized from the battery (except the battery cables) shall be marked using red characters. All other wires shall be marked using black characters. The use of colored wire coating as a marking scheme is prohibited. Wires, wire runs, and wiring harnesses shall be placed, bundled, and positioned so that access to other components is not inhibited or impeded. Any tying or lacing used to bundle individual wires into harnesses shall be of sufficient strength to hold the wires in position without crimping, slicing, or otherwise damaging wire insulation or strands. Wires, wire runs, and wiring harnesses shall be supported to retain position and provide support from stretching or sagging. Anti-chafe methods shall be employed to prevent damage from rubbing other components of the BEAR Freeze Protection Modules as well as abrasive damage due to sand/dust intrusion. Wires, wire runs, and wiring harnesses shall be protected from abrasive damage when entering, exiting, or passing through panels or bulkheads via the use of appropriately sized grommets.

3.15.3.1 Plugs and connectors. Weatherproof, sealed, O-ring connectors shall be used to the maximum extent possible to join wires and wiring harnesses and prevent the intrusion of water, sand, dust, condensation, salt fog, etc. into wiring connections. The use of military specification (MS) type plugs and receptacles is also encouraged. The use of terminal strips and ring or forked style crimp on connections for wire terminations shall be minimized. Unused openings in the plugs, receptacles, switches, or other similar wiring and control devices shall be sealed. Individual wires and wire runs within a wiring harness shall be the full length of the harness and not spliced together. The use of crimp-on 'piggyback' style splices used to join one or more wires together is prohibited.

3.15.4 FOD. Metal, rubber, and plastic tags, covers, or plugs commonly affixed to commercial items (see 6.3.1) during the manufacturing process shall be removed from the BEAR Freeze Protection Modules to prevent FOD hazards. All loose metal parts, such as pins, connector covers, caps, etc. shall be securely attached to the BEAR Freeze Protection Modules with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners.

3.15.5 Error-proof design. The design of the BEAR Freeze Protection Modules shall incorporate error-proofing in equipment mounting, installing, interchanging, connecting, and operating.

a. Equipment shall include physical features (for example, supports, guides, size, or shape differences, fastener locations, and alignment pins) that prevent improper mounting. In the absence of physical features, equipment shall be labeled or coded to identify proper mounting and alignment.

b. Equipment that has the same form and function shall be interchangeable throughout a system and related systems. If equipment is not interchangeable functionally, it shall not be interchangeable physically.

c. Connectors serving the same or similar functions shall be designed to preclude mismating or misalignment.

d. Design, location, procedural guidance, and suitable warning labels shall be provided to prevent damage to equipment while it is being handled, installed, operated, or maintained.

3.16 Workmanship. The BEAR Freeze Protection Modules, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship objectives shall include freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; marking of parts and assemblies; and proper alignment of parts and tightness of assembly fasteners.

3.16.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.16.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

3.16.3 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of excessive backlash.

3.16.4 Cleaning. The BEAR Freeze Protection Modules shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slag; stray bolts, nuts, and washers; rust; metal particles; pipe compound; and other foreign matter shall be removed during and after final assembly.

3.16.5 Touch-up. Any minor dings, scuffs, or scratches that may occur during the assembly process shall be cleaned, scuffed (as required), primed (thin tie coat), and painted in accordance with the corrosion prevention requirements stated herein, using the same products as applied to the surface(s) prior to assembly.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. BEAR Freeze Protection Modules inspection (see 4.2).

b. Operational inspection (see 4.3).

c. Conformance inspection (see 4.4).

Requirements shall be verified in accordance with Table IIII.

Table IIII. Requirement verification matrix.

Section 3 Requirement	Verification Method	Section 4 Verification
3.3 BEAR Freeze Protection Modules.	Examination	4.6.1 Examination of product.
3.4 Performance.	Examination	4.6.2 Performance tests.
3.4.4 Freeze protection components.	Examination	4.6.1 Examination of product.
3.5 Environmental conditions.	Test	4.6.5 Environmental testing.
3.5.1 Operating temperature range.	Test	4.6.5.1 Low temperature storage and operation test.
3.5.1 Operating temperature range.	Test	4.6.5.2 High temperature storage and operation test.
3.5.2 Storage temperature range.	Test	4.6.5.1 Low temperature storage and operation test.
3.5.2 Storage temperature range.	Test	4.6.5.2 High temperature storage and operation test.
3.5.3 Precipitation.	N/A	4.6.5.3 Precipitation tests and analysis.
3.5.3.1 Rain.	Test	4.6.5.3.1 Rain test.
3.5.3.2 Snow.	Analysis	4.6.5.3.2 Snow load analysis.
3.5.3.3 Ice.	Test	4.6.5.3.3 Ice accretion test.
3.5.4 Solar radiation.	Examination	4.6.1 Examination of product.
3.5.5 Fungus.	Examination	4.6.1 Examination of product.
3.5.6 Salt fog.	Test	4.6.5.4 Salt fog test.
3.5.7 Sand and dust.	Test	4.6.5.5 Sand and dust test.
3.5.8 Humidity.	Test	4.6.5.6 Humidity test.
3.6 Weight and dimensions.	N/A	4.6.6 Weight and dimension tests.
3.6 Weight and dimensions.	Test	4.6.6.1 Weight test.
3.6 Weight and dimensions.	Examination	4.6.6.2 Dimension measurement.

Section 3 Requirement	Verification Method	Section 4 Verification
3.7 Packaging, Marking and Transportability.	Analysis	4.6.7 Transportability verification.
3.8.1 Maintainability and service life.	N/A	4.6.8 Maintainability analysis and demonstration.
3.8.1.1 Preventive maintenance.	Demonstration	4.6.8.1 Preventive maintenance demonstration.
3.8.2 Service life.	Analysis	4.6.9 Service life analysis.
3.9 Human systems integration (HSI).	Examination	4.6.1 Examination of product.
~~3.10 Removed~~		
3.11.1 System safety.	Analysis	4.6.12 System safety hazard analysis.
3.11.2 Hazardous materials.	Examination	4.6.1 Examination of product.
3.11.3 Component protection.	Examination	4.6.1 Examination of product.
3.12 Protective coatings, finish, and corrosion control.	Examination	4.6.1 Examination of product.
3.12.1 Protective coatings.	Examination	4.6.1 Examination of product.
3.12.2 Surface preparation.	Examination	4.6.1 Examination of product.
3.12.2.1 Aluminum pretreatment.	Examination	4.6.1 Examination of product.
3.12.2.2 Ferrous pretreatment.	Examination	4.6.1 Examination of product.
3.12.3 Primer.	Examination	4.6.1 Examination of product.
3.12.3.1 Ferrous surfaces.	Examination	4.6.1 Examination of product.
3.12.3.2 Aluminum and mixed aluminum and ferrous surfaces.	Examination	4.6.1 Examination of product.
3.12.4 Topcoat.	Examination	4.6.1 Examination of product.
3.12.5 Finish.	Examination	4.6.1 Examination of product.
3.12.6 Exclusion of water.	Examination	4.6.1 Examination of product.
3.12.7 Fluid traps and faying surfaces.	Examination	4.6.1 Examination of product.
3.12.8 Ventilation.	Examination	4.6.1 Examination of product.
3.12.9 Drainage.	Examination	4.6.1 Examination of product.
3.12.10 Dissimilar metals.	Examination	4.6.1 Examination of product.
3.13 Markings.	Examination	4.6.1 Examination of product.
3.14.1 Identification plate.	Examination	4.6.1 Examination of product.
3.14.2 Unique Identification (UID).	Examination	4.6.1 Examination of product.
3.15 Design and construction.	Examination	4.6.1 Examination of product.
3.15.1 ~~REMOVED~~		
3.15.2 Mechanical design.	Examination	4.6.1 Examination of product.
3.15.2.1 Fastening devices.	Examination	4.6.1 Examination of product.

Section 3 Requirement	Verification Method	Section 4 Verification
3.15.2.2 Welders and welding.	Examination	4.6.1 Examination of product.
3.15.3 Wiring.	Examination	4.6.1 Examination of product.
3.15.3.1 Plugs and connectors.	Examination	4.6.1 Examination of product.
3.15.4 FOD.	Examination	4.6.1 Examination of product.
3.15.5 Error-proof design.	Examination	4.6.1 Examination of product.
3.16 Workmanship.	Examination	4.6.1 Examination of product.
3.16.1 Bolted connections.	Examination	4.6.1 Examination of product.
3.16.2 Riveted connections.	Examination	4.6.1 Examination of product.
3.16.3 Gear and lever assemblies.	Examination	4.6.1 Examination of product.
3.16.4 Cleaning.	Examination	4.6.1 Examination of product.
3.16.5 Touch-up.	Examination	4.6.1 Examination of product.
3.3 BEAR Freeze Protection Modules.	Test	4.6.13 Compatibility test.

Revise the list of examinations, tests, and analyses (in the first sentence) as appropriate. Delete the references to the compatibility test if it does not apply.

4.2 First article inspection. Four first article BEAR freeze protection universal and unique components as described in section 3.4.4 shall be subjected to the analyses, demonstrations, examinations, and tests described in 4.6.1 through 4.6.13. The contractor shall provide or arrange for all test equipment and facilities, except for the compatibility test described in 4.6.13. Unless otherwise approved by the procuring activity, all BEAR freeze protection components shall be in the same configuration at all times and configuration changes shall not be made during the BEAR freeze protection components inspection.

4.3 Operational inspection. All operational inspection shall be satisfactorily completed and approved prior to final BEAR freeze protection component acceptance.

4.4 Conformance inspection. Each production BEAR freeze protection component shall be subjected to the examination described in 4.6.1.

4.5 Inspection requirements.

4.5.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

4.5.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Ambient humidity.

4.5.3 Test rejection criteria. Throughout all tests specified herein, the BEAR freeze protection component shall be closely observed for the following conditions, which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Thermal or structural failure of any component, including permanent deformation, or evidence of impending failure.

c. Evidence of excessive wear. If excessive wear is suspected, the original equipment manufacturer's (OEM's) specifications or tolerances shall be utilized for making a determination.

d. Evidence of corrosion or deterioration.

e. Misalignment of components.

f. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

g. Interference between the BEAR freeze protection components.

4.5.4 Failure reporting, analysis, and corrective action system (FRACAS). A closed loop system shall be used to collect data, analyze, and record timely corrective action for all potential failures that occur during the first article inspection and operational tests. The contractor's existing FRACAS shall be utilized with the minimum changes necessary to conform to this specification. The system shall cover all test samples, interfaces between test samples, test instrumentation, test facilities, test procedures, test personnel, and the handling and operating instructions. The contractor shall establish and maintain a FRACAS database; all information shall be entered into the database. Authorized procuring activity personnel shall have read-only access to the FRACAS database.

4.5.4.1 Problem and failure action. At the occurrence of a problem or potential failure that affects satisfactory operation of a test sample, entries shall be made in the appropriate data logs and the failed test sample shall be removed from test, with minimum interruption to the other test samples continuing test.

4.5.4.1.1 Problem and failure reporting. A failure report shall be initiated at the occurrence of each problem or potential failure of the contractor hardware or software or Government furnished equipment (GFE). The report shall contain the information required to permit determination of the origin and correction of the failure. The contractor's existing failure report form may be used with minimum changes necessary to conform to the requirements of this specification. The failure report form shall include the information specified in a through c:

a. Descriptions of failure symptoms, conditions surrounding the failure, failed hardware identification, and operating time (or cycles) at the time of failure.

b. Information on each independent and dependent failure and the extent of confirmation of the failure symptoms, the identification of failure modes, and a description of all repair actions taken to return the test sample to operational readiness.

c. Information describing the results of the investigation, the analysis of all part failures, an analysis of the system design, and the corrective action taken to prevent failure recurrence. If no corrective action is taken, the rationale for this decision shall be recorded.

4.5.4.1.2 Identification and control of failed items. A failure tag shall be affixed to the failed part immediately upon the detection of any failure or suspected failure. The failure tag shall provide space for the failure report serial number and for other pertinent entries from the test sample failure record. All failed parts shall be marked conspicuously or tagged and controlled to ensure disposal in accordance with contract requirements. Failed parts shall not be handled in any manner which may obliterate facts which might be pertinent to the analysis. Failed parts shall be stored pending disposition by the authorized approval agency of the failure analysis.

4.5.4.1.3 Problem and failure investigations. An investigation and analysis of each reported failure shall be performed. Investigation and analysis shall be conducted to the level of hardware or software necessary to identify causes, mechanisms, and potential effects of the failure. Any applicable method (for example, test, microscopic analysis, applications study, dissection, X-ray analysis, spectrographic analysis) of investigation and analysis which may be needed to determine failure cause shall be used. When the removed part is not defective or the cause of failure is external to the part, the analysis shall be extended to include the circuit, higher hardware assembly, test procedures, and subsystem if necessary. Investigation and analysis of GFE failures shall be limited to verifying that the GFE failure was not the result of the contractor's hardware, software, or procedures. This determination shall be documented for notification of the procuring activity.

4.5.4.1.4 Failure verification. Reported failures shall be verified as actual failures or an acceptable explanation provided to the procuring activity for lack of failure verification. Failure verification is determined either by repeating the failure mode of the reported part or by physical or electrical evidence of failure (for example, leakage residue or damaged hardware). Lack of failure verification, by itself, is not sufficient rationale to conclude the absence of a failure.

4.5.4.1.5 Corrective action. When the cause of failure has been determined, a corrective action shall be developed to eliminate or reduce the recurrence of the failure. Repairs shall be made in accordance with normal field operating procedures and manuals. The procuring activity shall review the corrective actions at the scheduled test status review prior to implementation. In all cases the failure analysis and the resulting corrective actions shall be documented.

4.5.4.1.6 Problem and failure tracking and closeout. The closed loop failure reporting system shall include provisions for tracking problems, failures, analyses, and corrective actions. Status of corrective actions for all problems and failures shall be reviewed at scheduled test status reviews. Problem and failure closeout shall be reviewed to assure their adequacy.

4.6 Test methods.

4.6.1 Examination of product. Each BEAR freeze protection component shall be examined to verify compliance with the requirements herein prior to accomplishing any other demonstrations or tests listed in 4.6. A contractor-generated, Government-approved checklist (part of the test procedure) shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, fastening, and markings. Proper operation of each BEAR freeze protection component function shall be verified. Each production BEAR freeze protection component shall be inspected to a Government-approved reduced version of the checklist.

a. 3.5.4 Contractor certification that the BEAR Freeze Protection Modules components performance is not adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments.

b. 3.5.5 Contractor certification that the materials used in construction of the BEAR Freeze Protection Modules components are fungus resistant or suitably treated to resist fungus.

4.6.2 Performance tests.

4.6.2.1 Fault Identification. The contractor shall certify compliance with the requirements of 3.3 by demonstrating the ability to quickly identify faulty BEAR Freeze Protection Modules electrical and mechanical components in an operational configuration through visual indication.

4.6.2.2 Fault Tolerance. The contractor shall certify compliance with the requirements of 3.3 by demonstrating the ability of the BEAR Freeze Protection Modules to maintain operational requirements of 3.5.1 in the event of BEAR freeze protection component failure. An engineering analysis shall be performed to identify the required repair times of components that cannot withstand prolonged operation with known faults in the operational conditions described in 3.5.1.

4.6.2.3 Water System Freeze Remediation. The contractor shall certify compliance with the requirements of 3.3 by demonstrating the ability to return BEAR water system components to normal operating conditions due to failed freeze protection.

4.6.2.4 Liquid Water Draw from Frozen Sources. An engineering analysis shall be performed to demonstrate compliance with the ice penetration requirement of 3.3.

4.6.2.5 BEAR Water System Compatibility. The contractor shall certify compliance with the requirements of 3.3 by demonstrating the compatibility of the freeze protection system with the current configuration BEAR water system without modification.

4.6.2.6 BEAR Freeze Protection Modularity, Scalability, and Configuration Flexibility. The contractor shall certify compliance with the requirements of 3.3 by demonstrating the ability to configure the BEAR Freeze Protection Modules to be easily adapted to different operational needs.

4.6.2.7 BEAR Freeze Protection Installation. The contractor shall certify compliance with the requirements of 3.4.1 by demonstrating the ability to install the BEAR Freeze Protection Modules to a fielded water system configured as described by the procuring activity.

4.6.2.8 BEAR Freeze Protection Electrical. The contractor shall certify compliance with the requirements of 3.4.2 by demonstrating the ability to operate the BEAR Freeze Protection Modules with the current configuration BEAR electrical distribution equipment.

4.6.2.8 BEAR Water System Configuration and Layout Deviation Verification. The contractor shall certify compliance with the requirements of 3.4.3 by demonstrating the ability to reconfigure BEAR water system components without affecting overall system operation.

4.6.2.9 BEAR Freeze Protection Equipment. The contractor shall certify compliance with the requirements of 3.4.4 through 3.4.4.10 by installation and operational demonstration.

4.6.5 Environmental testing.

4.6.5.1 Low temperature storage and operation test. BEAR Freeze Protection Modules components shall be tested in accordance with MIL-STD-810, Method 502.7, Procedures I and II, to demonstrate compliance with the low temperature storage and operating requirements of 3.5.1 and 3.5.2. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

4.6.5.2 High temperature storage and operation test. BEAR Freeze Protection Modules components shall be tested in accordance with MIL-STD-810, Method 501.7, Procedures I and II, to demonstrate compliance with the high temperature storage and operating requirements of 3.5.1 and 3.5.2. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

4.6.5.3 Precipitation tests and analysis.

4.6.5.3.1 Rain test. BEAR Freeze Protection Modules components shall be tested in accordance with MIL-STD-810, Method 506.6, Procedure I, to demonstrate compliance with 3.5.3.1.

4.6.5.3.2 Snow load analysis. An engineering analysis shall be performed to demonstrate compliance with the snow load requirement of 3.5.3.2, using a specific gravity of snow of 0.1 (Ref. 5.3 of MIL-STD-810, Part Three).

4.6.5.3.3 Ice accretion test. BEAR Freeze Protection Modules components shall be tested in accordance with MIL-STD-810, Method 521.4 with an ice thickness of 1.5-inches to demonstrate compliance with the ice accretion requirement of 3.5.3.3. The contractor shall identify those areas of the BEAR Freeze Protection Modules components where ice removal is required prior to operation.

4.6.5.4 Salt fog test. BEAR Freeze Protection Modules components shall be tested in accordance with MIL-STD-810, Method 509.7, to demonstrate compliance with 3.5.6. Test duration shall be alternating 24-hour periods of salt fog exposure and drying conditions for 24-hour periods (two wet and two dry).

4.6.5.5 Sand and dust test. BEAR Freeze Protection Modules components shall be tested in accordance with MIL-STD-810, Method 510.7, Procedures I (12 hours) and II (90 minutes per side), to demonstrate compliance with 3.5.7.

4.6.5.6 Humidity test. BEAR Freeze Protection Modules components shall be tested in accordance with MIL-STD-810, Method 507.6, Procedure Ib, to demonstrate compliance with 3.5.8.

4.6.6 Weight and dimension tests.

4.6.6.1 Weight test. The weight of the BEAR Freeze Protection Modules modules shall be measured to demonstrate compliance with the weight requirement of 3.6.1.

4.6.6.2 Dimension measurement. The BEAR Freeze Protection Modules modules shall be measured to demonstrate compliance with the dimensional requirements of 3.6.1.

4.6.7 Transportability verification. An engineering analysis shall be performed to demonstrate compliance with 3.7.

4.6.8 Maintainability analysis and demonstration.

4.6.8.1 Preventive maintenance demonstration. All recommended preventive maintenance tasks shall be performed and the task times shall be recorded. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472. All preventive maintenance tasks recommended to be performed daily and at the routine PMI shall also be performed by personnel wearing arctic mittens.

4.6.9 Service life analysis. An engineering analysis shall be performed to demonstrate compliance with the service life requirement of 3.8.2.

4.6.12 System safety hazard analysis. A system safety hazard analysis of the {ITEM} shall be conducted in accordance with 4.3.1 through 4.3.8 of MIL-STD-882 to demonstrate compliance with the mishap risk requirement of 3.12.1.

4.6.13 Compatibility test. The contractor shall certify compliance with the requirements of 3.3 by demonstrating the compatibility of the freeze protection system with the current configuration BEAR water system, and electrical distribution equipment without modification.

5. PACKAGING

5.1 For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The BEAR Water Freeze Protection System is an additive package to be used with the BEAR Water System to provide water distribution capabilities in temperature ranges from 32°F to -25°F. The BEAR Water Freeze Protection System does not provide the capability to meet base water requirements alone.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this BEAR Water Freeze Protection Purchase Description.

b. If BEAR Freeze Protection Modules component inspection is required (see 3.1).

c. Packaging requirements (see 5.1).

d. Specification of finish color (see 3.12.5)

Appendix I

BEAR Water System Major Components

		
Diesel Driven Pump Assembly, Trailer Mounted, 400 GPM	**20,000 Gallon Collapsible Fabric Tank**	**3,000 Gallon Collapsible Fabric Tank**
Dimensions: 97" x 87" x 77 ¾" .	Dimensions: 24' X 28' x 60"	Dimensions: 14' X 14" x 38"
Weight: 3,200 lbs	Weight: 1,120 lbs. (empty)	Weight: 115lbs. (empty)
National Stock Number: 4320-01-551-9776 Part Number: 3010220A	National Stock Number: 5430-01-602-4521 / 4507 / 4524 Part Number: 3010100-1 / 2 / 3	National Stock Number: 5 430-01-562-1281 / 5430-01-559-0016 Part Number: 3020200-1 / 3020300-4
Technical Order Number: TO 40W4-21-11	Technical Order Number: 40W4-21-11	Technical Order Number: 40W4-21-11
Configured in: SRS (FP Module 1)	Configured in: WPS (FP Module 2) IDS (FP Module 5)	Configured in: WPS (FP Module 2) 550I (FP Module 3) 550F (FP Module 4) IDS (FP Module 5)

		
Reverse Osmosis Water Purification Unit (ROWPU) 1500 GPH	**35 GPM Electric Pump Assembly**	**Dual Pump Station**
Dimensions:104" x 88" x 84"	Dimensions: 27-1/2" X 31-3/4" X 25-3/4"	Dimensions: 84" x 60" x 74"
Weight: 8,000 lbs	Weight: 155 lbs.	Weight: 2,538 lbs.
National Stock Number: 4610-01-565-5686 Part Number: 701-1500	National Stock Number: 4320-01-558-6502 Part Number: 3020131-1 / 3	National Stock Number: 4540-01-558-8503 PN#: 3030140-1
Technical Order Number: TO 40W4-20-1	Technical Order Number: TO 40W4-21-1	Technical Order Number: TO 40W4-21-11
Configured in: WPS (FP Module 2)	Configured in: WPS (FP Module 2) IDS (FP Module 5)	Configured in: 550I (FP Module 3)

		
Macerator Pump Lift Station	**Dual Pump Lift Station**	**25,000 Gallon Waste Tank with Aeration System**
Dimensions: 36" x 84"	Dimensions: 60" X 60"	Dimensions: 31'x5'
Weight: 750 lbs.	Weight: 1,820 lbs.	Weight: 2,335 lbs. (empty)
National Stock Number: 4320-01-606-8080 Part Number: 3030190-3	National Stock Number: 4630-01-558-8536 Part Number:	National Stock Number: 4540-01-558-8503
Technical Order Number: TO 40W4-21-11	Technical Order Number: TO 40W4-21-11	Technical Order Number: TO 40W4-21-11
Configured in: WPS (FP Module 2)	Configured in: 550I (FP Module 3)	Configured in: 550I (FP Module 3)

		
Bladder Water Level Controller	**Kitchen Sewage Ejector Lift Station**	**Latrine Sewage Ejector System**
Dimensions: 20" X 20" X 20"	Dimensions: 30" X 22"	Dimensions: 36" X 30"
Weight: 66 lbs.	Weight: 102 lbs.	Weight: 130 lbs.
National Stock Number: 4820-01-558-8883 Part Number: 3030121-1	National Stock Number: 4630-01-558-8091 Part Number: 3040120	National Stock Number: 4630-01-558-9592 Part Number: 3030350-3
Technical Order Number: TO 40W4-21-11	Technical Order Number: 40W4-21-11	Technical Order Number: TO 40W4-21-11
Configured in: 550I (FP Module 3) 550F (FP Module 4) IDS (FP Module 5)	Configured in: 550F (FP Module 4) (Optional)	Configured in: 550I (FP Module 3) (Optional) 550F (FP Module 4) (Optional)

		
Pump Assy, Diesel, 125 GPM	**Banjo Style Ball Valve**	**Various Hoses & Fittings**
Dimensions: 22" X 22" X 26"	Various Sizes: 2", 3", 4"	Various Sizes: Inside Dia (in).: 1, 1-1/2, 2, 3, 4, 6 Length (ft): 5, 15, 25, 50, 100, 150
Weight: 149 lbs.		
National Stock Number: 4320-01-547-8734 Part Number: 3020220-1 / 2 / 3	Part Number: V200 (2"), V300 (3"), V400 (4")	
Technical Order Number: TO 40W4-21-11		Hoses: A-A-59566, A-A-59567 Fittings: A-A-59326 Class A Style 1 or 2, ASTM D2466, D2665
Configured in: WPS (FP Module 2) 550I (FP Module 3) 550F (FP Module 4)	Configured in all Modules	Configured in all Modules